قناة أولى تمريضيانو على التليجرام

FIRST YEAR

FUNDAMENTAL OF NURSIN

PRATICAL BOOK

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Hand hygiene and Handwashing

Objectives:

Students will be able to:

- Define of Hand Hygiene and Handwashing
- Differentiate between types of Hand Hygiene
- Determine equipment's needed during the procedure
- Demonstrate hand washing

Procedures:

- Antiseptic Hand rub
- Antiseptic Handwash
- Surgical Antisepsis

Terms

- **Hand Hygiene.** It is a general term that applies to handwashing, antiseptic handwash, antiseptic hand rub, or surgical hand antisepsis
- **Hand Washing.** It is defined as the washing of hands with plain (i.e., non-antimicrobial) soap and water.
- **Antiseptic Handwash.** A term that applies to handwashing with an antimicrobial soap and water.
- **Surgical Hand Antisepsis.** Commonly called as a surgical hand scrub. This is to remove as many microorganisms from the hands as possible before the sterile procedure.

Purposes

The purposes of hand hygiene are:

- Hand washing can prevent infection
- Avoid pathogenic microorganisms and to avoid transmitting them

Types of Hand Hygiene

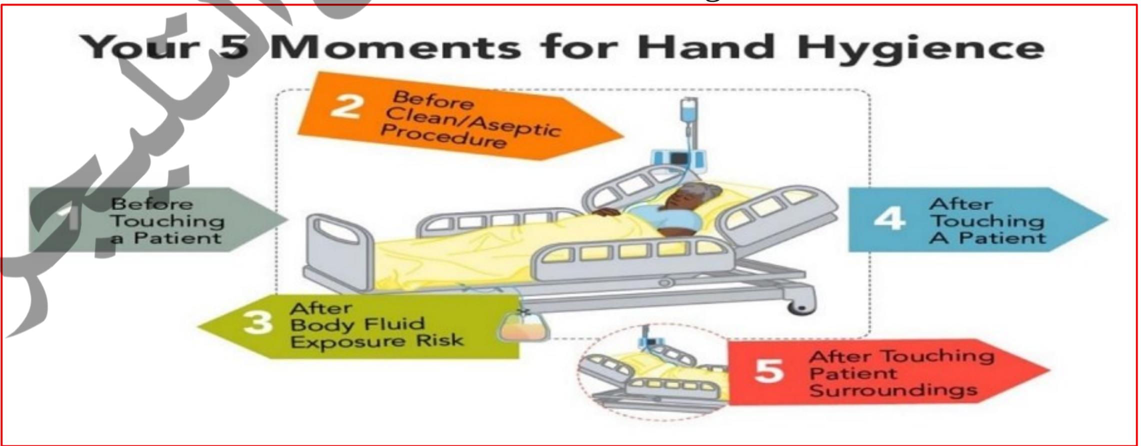
The following are the types of hand hygiene:

- **Routine handwash.** Use of water and non-antimicrobial soap for the purpose of removing soil and transient microorganisms.
- **Antiseptic handwash.** Use of water and antimicrobial soap (e.g., chlorhexidine, iodine and iodophors, chloroxylenol, triclosan) for the purpose of removing or destroying transient microorganisms and reduce resident flora.
- **Antiseptic hand rub.** Use of alcohol-based hand rub.
- **Surgical antisepsis.** Use of water and antimicrobial soap (e.g., chlorhexidine, iodine and iodophors, chloroxylenol, triclosan) for the purpose of removing or destroying transient microorganisms and reduce resident flora. Recommended duration is 2-6 minutes.

Indicators of Hand Hygiene

According to the World Health Organization (WHO), there are Five Moments for Hand Hygiene:

1. Before Patient Contact.
2. Before and Antiseptic Task.
3. After Body Fluid Exposure Risk.
4. After Patient Contact.
5. After Contact with Patient Surroundings.



Hand Hygiene Moments. Image via: WHO.int

Supplies Needed

The following equipment are needed to perform hand washing:

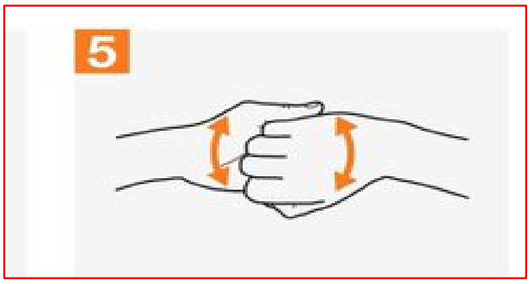
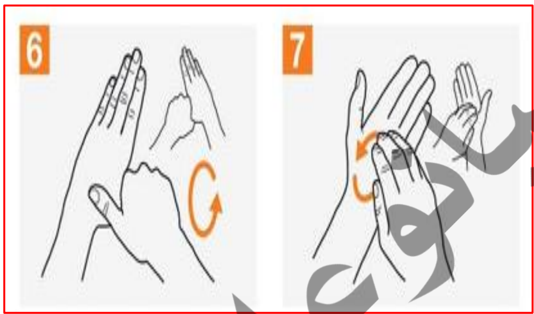
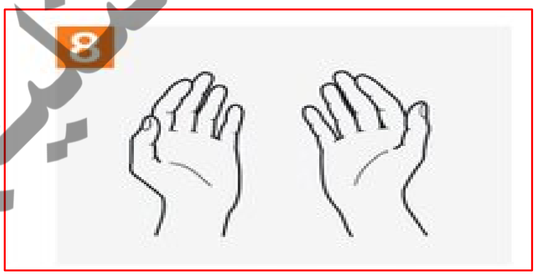
- Soap or detergent
- Warm running water
- Paper towels
- Alcohol
- Optional: Antiseptic cleaner, fingernail brush, plastic cuticle stick

Procedures

Below is the step-by-step guide for different hand hygiene methods:

Antiseptic Hand rub: The use of alcohol-based hand rub.

Steps	Rational
1. Ensure jewelers have been removed	To avoid microorganisms to accumulate inside accessory
2. Apply quantity of alcohol-based hand hygiene product as per manufacturer's recommendations into cupped hand. 	To ensure cover all surfaces of your hands and fingers
3. Rub hands palm to palm. Right palm over left dorsum with interlaced fingers and vice versa. 	To clean all surfaces

4. Backs of fingers to opposing palms with fingers interlaced. 	To be cleaned well
5. Rotational rubbing of left thumb clasped in right palm and vice versa. Rotational rubbing, backwards and forwards with clasped fingers of right hand in left palm and vice versa. 	To clean all fingers To clean under nails
6. Rubbing hands together until hands are dry before continuing with patient care, or put on gloves. 	

Antiseptic Hand wash

Also known as clean technique, includes procedures used to reduce the number of organisms on hands.

Steps	Rational
1.Prepare the necessary equipment (liquid or bar soap, paper towels, orange wood stick, lotion)	To save time and effort
2. Stand in front of sink. Don't allow your uniform to touch the sink during the washing.	To prevent cross infection
3. Remove jewelry & secure it in safe place.	To avoid microorganisms to accumulate inside accessory
4. Turn on water & adjust force & regulate the temperature until the water is warm.	To adjust water temperature
5. Wet the hands & wrist area. Keep hands lower than elbows to allow water to flow toward fingertips.	To avoid cross infection
6. Use enough amount of liquid soap (3-5ml) from dispenser or rinse bar of soap & leather thoroughly.	To cover all areas in hands
7.With firm rubbing and circular motions, wash the palms and backs of the hands, each finger, the knuckles, wrists, and forearms. Continue this friction motion for 30 seconds.	To ensure cover all surfaces
8. Use fingernail of other hand or clean orangewood sticks to clean under fingernail.	To clean under nails
9.Rinse thoroughly with water flowing towards the fingertips	
10. Dry hands, beginning with fingers & moving upward toward forearm, with a paper towel & discard it immediately without touching other clean hands.	

Surgical hand scrub

Also known as sterile technique, prevents contamination of an open wound, serves to isolate the operative area from the unsterile environment, and maintains a sterile field for **surgery**

Steps	Rational
1. Remove all pieces of jewelry.	To avoid microorganisms to accumulate inside accessory
2. Wet hands using sterile water with water closest to your body temperature.	To adjust water temperature
3. Wash hands to elbow using antimicrobial soap and/or povidone-iodine.  	To remove microorganisms
4. Clean subungual areas with a nail file 	To remove microorganisms

5.Scrub each side of each finger, between the fingers, and the backs and fronts of the hands for at least 4 minutes.	
6. Proceed to scrub the hands, keeping the hand higher than the arm at all times. 	to prevent bacteria-laden soap and water from contaminating the hands.
7. Rinse hands and arms by passing them through the flowing water in one direction only, from fingertips to elbow. 	To be cleaned
8. Proceed to the operating room holding hands above elbows.	
9. Dry hands and arms using sterile towel observing aseptic technique. 	

Checklist for different types of hand hygiene

Checklist for hand rub

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1.Ensure jewelers removed					
2.Apply alcohol-based hand hygiene product.					
3.Rub hands palm to palm.					
4.Backs of fingers to opposing palms with fingers interlaced.					
5.Rotational rubbing of the thumb clasped in palm. Rotational rubbing backwards and forwards with clasped fingers of the hand in palm.					
6.Rubbing hands together until hands are dry before put on gloves.					

Checklist for hand wash

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Prepare the necessary equipment.					
2. Remove jewelry.					
3.Stand in front of sink.					
4.Turn on water & adjust temperature.					
5.Wet the hands & wrist area.					
6.Use enough amount of liquid soap & leather thoroughly.					
7.With firm rubbing and circular motions, wash the palms, hands back, each finger, wrists, and forearms. Continue this friction motion for 30 seconds.					
8.Use fingernail of other hand					
9.Rinse thoroughly with water.					
10.Dry hands & discard it immediately.					

Checklist surgical hand scrub

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Remove all pieces of jewelry.					
2. Wet hands using sterile water.					
3. Wash hands to elbow using antimicrobial soap.					
4. Clean subungual areas with a nail file.					
5. Scrub each side of each finger, between the fingers, and the backs and fronts of the hands for at least 4 minutes.					
6. Rinse hands and arms by passing them through the flowing water in one direction only, from fingertips to elbow.					
7. Proceed to the operating room holding hands above elbows.					
8. Dry hands and arms using sterile towel.					

Personal Protective Equipment (PPE)

Objectives:

Students will be able to:

- Define of Personal protective equipment
- Describe the types of Personal protective equipment.
- Determine equipment's needed during the procedure
- Apply Personal protective equipment.

Definition:

Personal protective equipment (PPE) is a specialized clothing or equipment worn by healthcare providers for protection against health effects & safety hazards. It is designed to protect many parts ie. Eyes, head, face, hand, feet, ear.

Purposes:

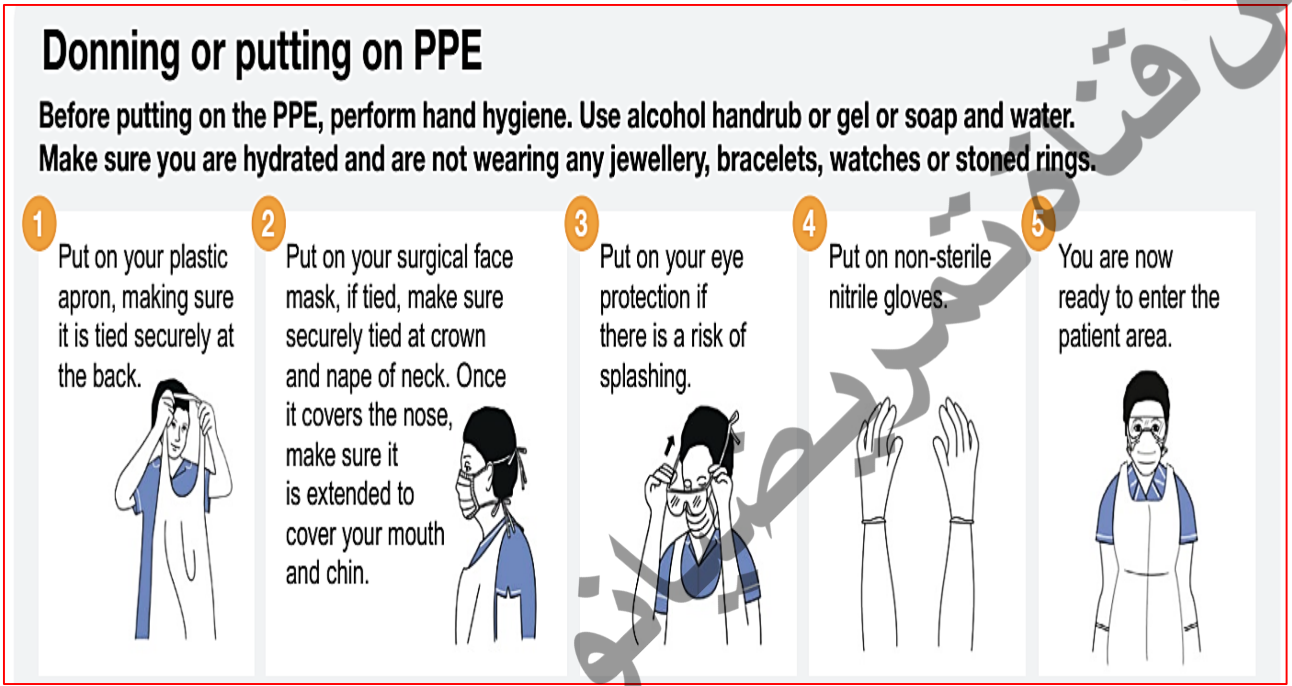
- To enhance easy handling of sterile equipment's.
- To reduce the risk of transmission of microorganism to patient.
- To reduce the risk of transmission of infectious agent to oneself.
- To prevent cross infections.
- To prevent wound infection post operatively.
- To prevent contamination of sterile field.

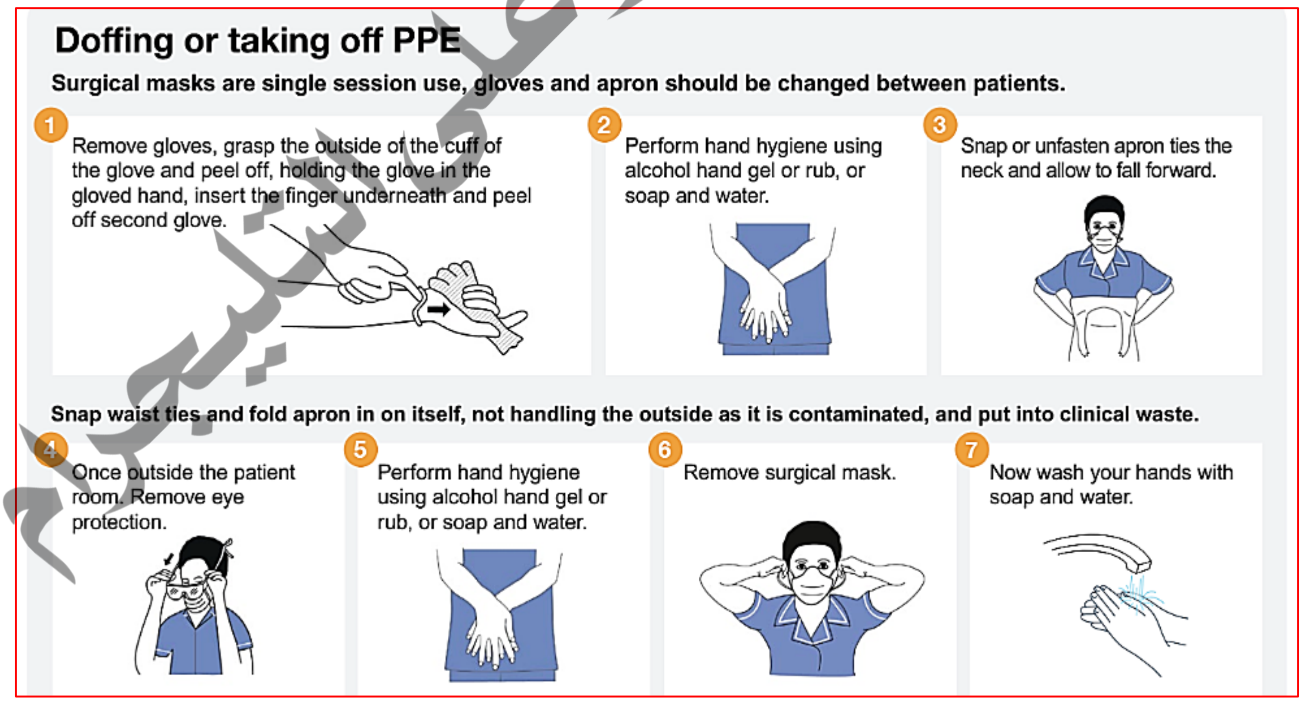
Types of PPE:

- Foot protection (overshoes).
- Head protection (cap or overhead).
- Gowns/aprons – protect skin and/or clothing.
- Masks and respirators– protect mouth/nose & protect respiratory tract from airborne infectious agents.

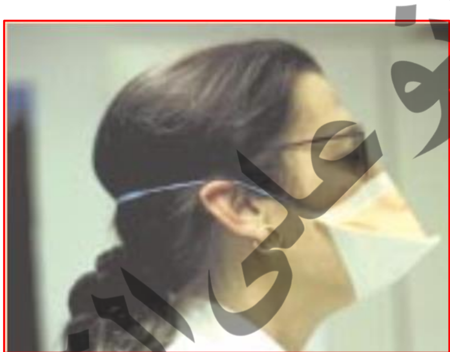
- Goggles – protect eyes.
- Face shields – protect face, mouth, nose, and eyes.
- Gloves – protect hands.

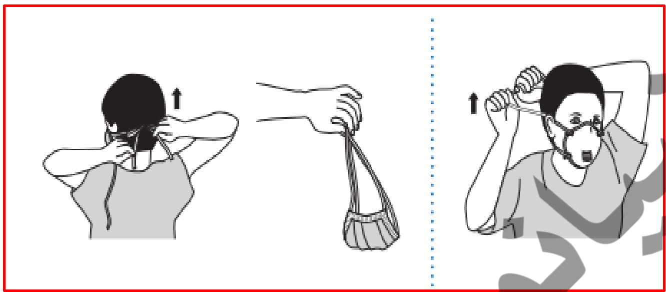
The following is a demonstration of the full order of donning PPE:





Donning and doffing cap & mask

Steps	Rational
1. Wash hands	Reduces transmission of microorganisms.
2. Apply cap to head, being sure to tuck hair under cap. Males with facial hair should use a hood to cover all hair on head and face. 	prevent the transmission of organisms from the nurse to the client. The cap also protects the nurse from infectious pathogens
3. Secure mask around mouth and nose  For masks with strings: a. Hold mask by top and pinch metal strip over bridge of nose. b. Pull two top strings over ears and tie at upper back of head. c. Tie two lower ties around back of neck so that bottom of mask fits snugly under chin.	

4. Enter the client's room and explain the rationale for wearing a cap and mask.	Minimizes anxiety and feelings of isolation
Doffing of Cap & Mask	
1. After performing necessary tasks, remove cap and mask before leaving room.	Reduces transmission of organisms.
2. Untie bottom strings of mask first, then top strings, and lift off of face. Hold mask by strings and discard. 	Prevents contaminated surface of mask from contacting uniform.
3. Grasp top surface of cap and lift from head	Minimizes contact of hands to hair
4. Wash hands after removing mask	Reduces transmission of microorganisms.
5. Document the type of protective barriers used	

Checklist Donning & Doffing of Gown

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
Wash hands					
Apply cap to head.					
3.Secure mask around mouth and nose For masks with strings: a. Hold mask by top and pinch metal strip over bridge of nose. b. Pull two top strings over ears and tie. c. Tie two lower ties around back of neck so that bottom of mask fits snugly under chin.					
Doffing of Cap & Mask					
Remove cap and mask before leaving room.					
Untie strings of mask, and lift off of face. Hold mask by strings and discard.					
Grasp top surface of cap and lift from head					
Wash hands.					
Document the type of protective barriers used					

Donning & Doffing gown

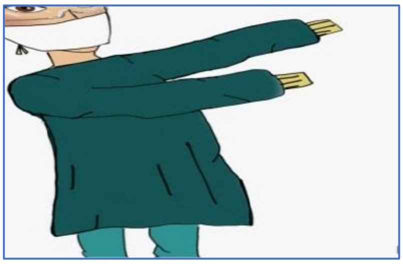
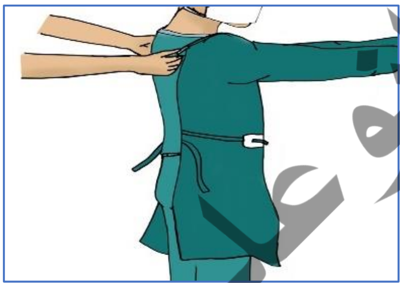
Gowns or aprons

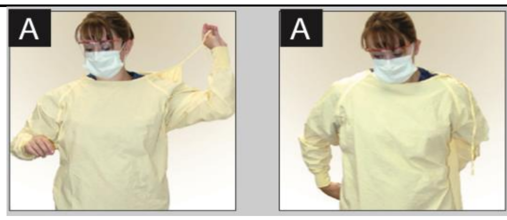
Providing an effective barrier should be worn during invasive procedures likely to result in the splashing of blood or other body fluids.

Purpose:

1- To prevents spread of microorganisms from the nurse to the patient
2- To prevents contamination of the nurse clothing from the patient.
3- To prevent spread of air borne microorganisms from the patient to the nurse

Steps	Rationale
1- Wash your hands.	To prevent cross infection
2- Opens wrapped gown package.	To be ready
4- Pick up the gown. 	
4-Unfolds the gown while holding the inner neck area. 	To keep sterile
5-Insert each arm in the gown.	

	
6-Mack sure the gown completely covers the front of your uniform. 	Gown should protect entire uniform.
7-Tie the strings at the back of the neck. 	
8-Remove gown:	
A-Gown that is not visibly soiled requires no particular technique for removal.	
For gown that is visibly soiled: -- b-Untie neck strings of gown. Remove gown without touching outside of gown by keeping one hand up and under the	- Neck strings are considered clean. Outside of gown is contaminated.

gown cuff and using this protected hand to pull the opposite sleeve down and off.	
C-Use ungowned arm and hand to grasp the gown from the inside and remove from the remaining arm. Remove gown and turn inside out and drop in appropriate container.	To prevent cross infection. 
9-Wash hands thoroughly.	

Checklist Donning & Doffing gown

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1- Wash your hands.					
2- Opens wrapped gown package.					
5- Pick up the gown.					
4-Unfolds the gown while holding the inner neck area.					
5-Insert each arm in the gown.					
6-Mack sure the gown completely covers the front of your uniform.					
7-Tie the strings at the back of the neck.					
8-Remove gown:					
A-Gown that is not visibly soiled requires no particular technique for removal.					
For gown that is visibly soiled: b-Untie neck strings of gown. Remove gown without touching outside of gown then pull the opposite sleeve down and off.					
C-. Remove gown and turn inside out and drop in appropriate container.					
9- Wash hands thoroughly.					

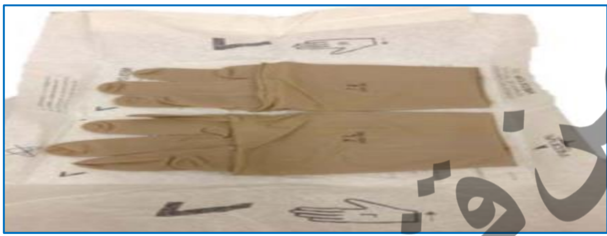
Donning & doffing gloving

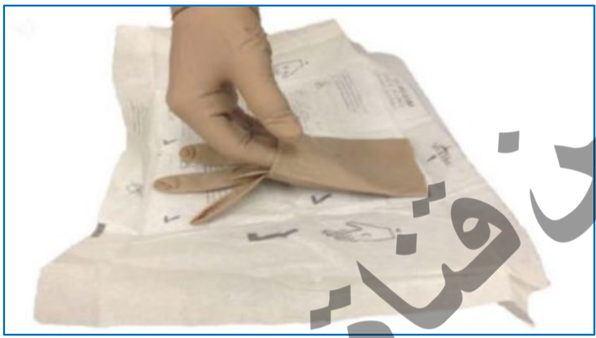
Gloves

Must be worn for all in invasive procedures.

Donning sterile gloves

Steps	Rationale
1-Wash and dry hands carefully.	Hand washing deters the spread of microorganisms. Gloves are easier to don when hands are dry.
2- Choose the correct size glove.	Glove must be fit
3- Place sterile glove package on clean, dry surface above your waist.	Moisture could contaminate the sterile gloves. Any sterile object held below the waist is considered contaminated
4-Open the outside wrapper by carefully peeling the top layer back. Remove inner package, handing only the outside of it.	This maintains sterility of gloves in inner packet. 

5- Carefully open the inner package and expose the sterile gloves with the cuff end closest to you.	-The inner surface of the package is considered sterile. 
6- With the thumb and forefinger of non-dominant hand, grasp the top edge of the folded cuff of the sterile glove for dominant hand.	-Unsterile hand only touches sterile of glove. Outside remains sterile. 
7- Life and hold glove with fingers down. Be careful it does not touch any unsterile object.	-Glove is contaminated if it touches un sterile objects. 
8- Carefully insert the dominant hand glove and pull glove on. Leave cuff folded down until other hand is gloved.	-Attempts to turn upward with unsterile hand may result in contamination of sterile glove

9- Holding thumb outward, slide fingers of gloved hand under cuff of remaining glove and lift glove upward.	-Thumb is less likely to become contaminated if held outward 
10- Carefully insert non-dominant hand into glove. Adjust gloves on both hands touching only sterile areas.	-Sterile surface touching sterile surface prevents contamination. 

Doffing sterile gloves

Steps	Rationale
1-Using dominant gloved hand, grasp other glove near cuff end and remove by inverting it, keeping the contaminated area on the inside. Continue to hold on to glove.	- Contaminated area does not come in contact with hands or wrists. 
2- Slide fingers of ungloved hand inside the remaining glove. Grasp glove on inside and remove by turning inside out over hand and other glove.	- Contaminated area does not come in contact with hands or wrists. 
1- Discard gloves in appropriate container and wash hands. 	- Hand washing reduces the spread of microorganisms. 

Checklist Donning sterile gloves

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1-Wash and dry hands carefully.					
2- Choose the correct size glove.					
3- Place sterile glove package above your waist.					
4-Open the outside wrapper. Remove inner package, handing only the outside of it.					
5- Carefully open the inner package and expose the sterile gloves.					
6- With the thumb and forefinger of non-dominant hand, grasp the top edge of the folded cuff of the sterile glove for dominant hand.					
7- Life and hold glove with fingers down.					
8- Carefully insert the dominant hand glove and pull glove on.					
9- Holding thumb outward, slide fingers of gloved hand under cuff of remaining glove and lift glove upward.					
10- Carefully insert non-dominant hand into glove. Adjust gloves on both hands.					

Checklist Doffing sterile gloves

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1-Using dominant gloved hand, grasp the other glove near cuff end and remove by inverting it.					
2- Slide fingers of ungloved hand inside the remaining glove. Grasp glove on inside and remove.					
3-Discard gloves in appropriate container and wash hands.					

Vital signs

Objectives:

Students will be able to:

- Define vital signs.
- List the purpose of vital signs.
- Identify times to assess vital signs.

Definition:

Taking vital signs are defined as the procedure that takes the sign of basic physiology that includes temperature, pulse, respiration and blood pressure. If any abnormality occurs in the body, vital signs change immediately.

Purpose:

1. To assess the patient's condition
2. To determine the baseline values for future comparisons
3. To detect changes and abnormalities in the condition of the patient.

Times to assess vital signs:

1. Upon admission to any healthcare agency.
2. Based on agency institutional policy and procedures.
3. Any time there is a change in the patient's condition.
4. Before and after surgical or invasive diagnostic procedures.
5. Before and after activity that may increase risk.
6. Before and after administering medications that affect cardiovascular or respiratory functioning.
7. Before and after any nursing intervention that could affect the vital signs (e.g., ambulating a client who has been on bed rest).

1-Body temperature

Objectives:

Students will be able to:

- Define body temperature.
- Discuss the physiology of Body Temperature.
- Identify times to assess vital signs.
- Describe how to select the proper method to measure body temperature.
- Identify factors that affect body temperature.
- Explain alterations in Body Temperature.
- Differentiate between types of thermometers.
- Demonstrate measuring body temperature procedure.

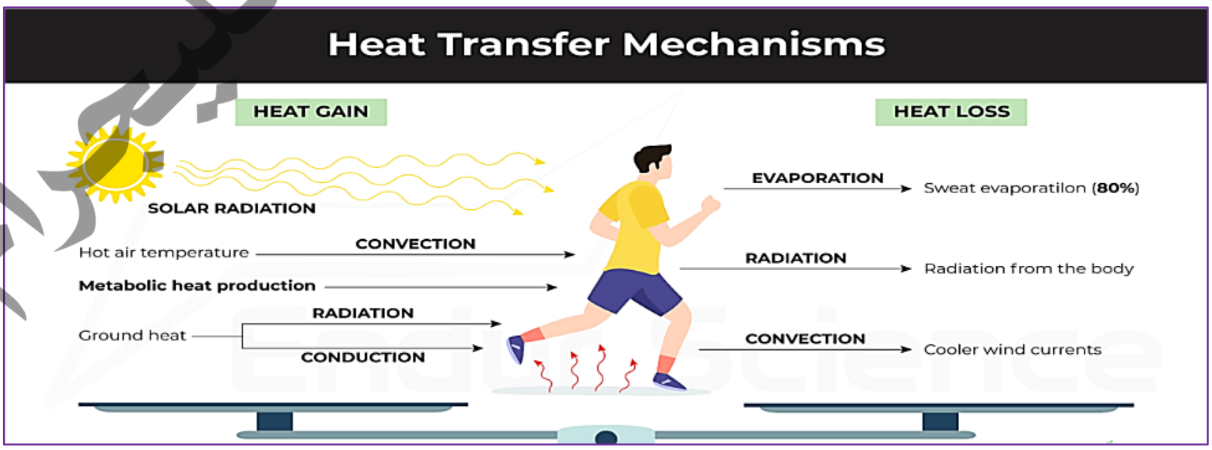
Definition:

Body temperature is the balance between heat produced in the tissues and heat lost to the environment.

The normal range of the body temperature is between 36.5-37.5c.

Physiology of Body Temperature:

Body temperature is the balance between the heat production due to chemical activities by the body and heat lost from the body through radiation, conduction, convection, and vaporization (evaporation).



Method for Measuring Body Temperature:

Method	Advisable	Inadvisable
Oral (“O”)	Most adults and children who are able to follow instructions for properly holding the thermometer.	Patients who have had oral surgery, mouth sores, dyspnea; uncooperative patients; patients on oxygen; infants and small children
Rectal (“R”)	Infants and small children; patients who have had oral surgery; mouth-breathing Patients; unconscious patients.	Active children; those with recent rectal surgery or complaints of diarrhea.
Axillary (“AX”)	Small children	Patients who have underarm rash, excessive perspiration.
Tympanic (Aural) (“T”)	Small children	Patient with in-the-ear hearing aids, impacted cerumen, earaches, or ear infections.
Temporal Artery (“TA”)	Most adults, infants, and small children; patients who have had oral surgery; mouth-breathing patients; unconscious patients.	No restrictions—possibly difficult with combative children or newborns.

Factors Affecting Body Temperature	
Time of Day	Body temperature is lower in the morning upon waking, when metabolism is still slow, and the body's highest temperature usually occurs in the evening.
Age	Infants and children normally have a higher body temperature than adults.
Gender	Women may experience a slight increase in body temperature at the time of ovulation.
Physical exercise	Body temperature will rise during exercise.
Emotions	Emotions such as crying and anger can cause an increase in body temperature.
Pregnancy	An increase in metabolism during pregnancy may cause the body temperature to rise.
Environmental Changes	Exposure to excessively hot/cold temperatures will increase/decrease body temperature.
Infection	An elevated temperature may be one of the first signs of an infection.
Drugs	Drugs may increase muscular activity or metabolism, which in turn increases temperature. Antipyretic lower the above-normal temperature.
Food	The process of eating and digestion may cause a rise in the body temperature.

Alterations in Body Temperature:

- **Fever or pyrexia:** is a body temperature above 38°C.

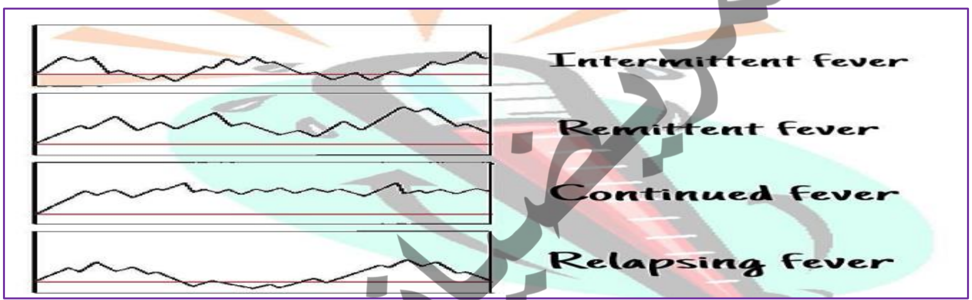
  The four most common types of fevers are:

1**. Intermittent Fever:** during this type of fever, the body temperature alternates at regular intervals between periods of fever and periods of normal temperatures.

2. Remittent Fever: during this type of fever, a wide range of temperature fluctuations occurs over the 2-hour period, all of which are above normal.

3. Relapsing Fever: In a relapsing fever, short febrile periods of a few days are interspersed with periods of 1 or 2 days of normal temperature.

4. Constant Fever: during a constant fever, the body temperature fluctuates minimally but always remains elevated.

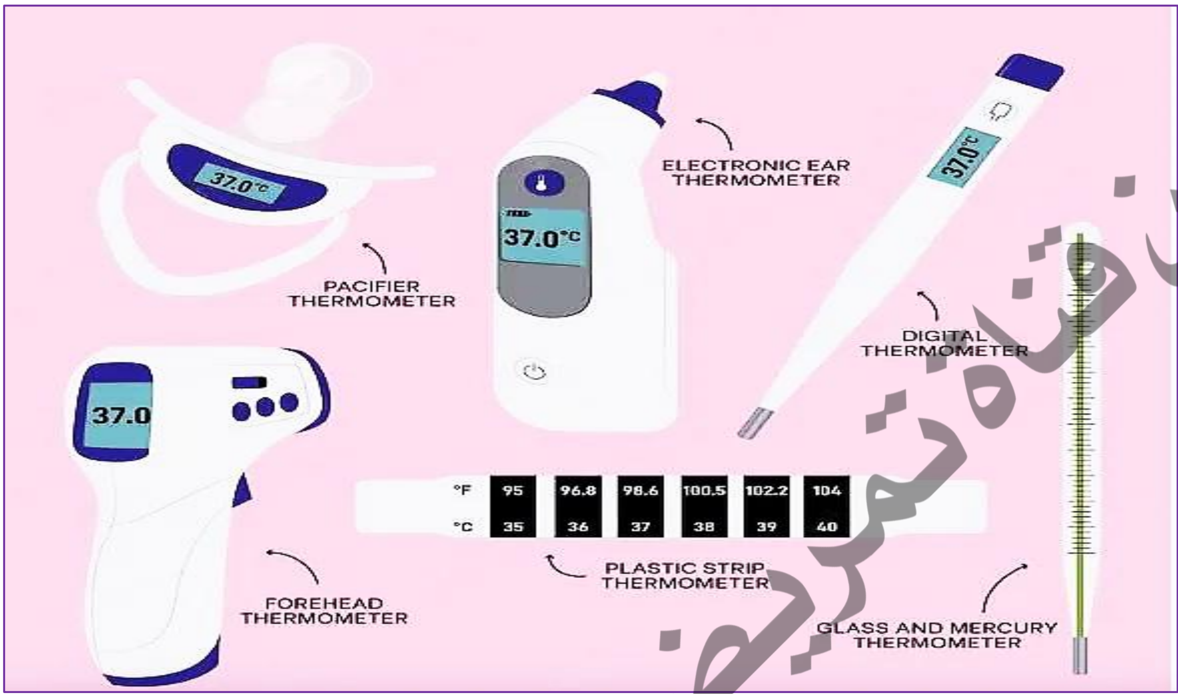


- **Hyperpyrexia or hyperthermia:** is a very high fever (41.5 °C) resulting from a regulated rise in core body temperature, usually a response to a physiological threat, such as an infection.
- **Hypothermia:** defined as a body temperature below (35°C) and is the result of the body losing more heat than it is producing.

Sites for Assessing Temperature:

1. Orally (common way). (3 – 5 min).
2. Axillary (safe way). + 0.5 °C (10 min).
3. Rectal (accurate reading). – 0.5 °C (2 – 3 min).
4. Tympanic membrane.

Types of thermometers:



Measuring body temperature

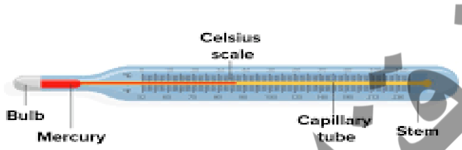
Equipment:

- Appropriate thermometer
- Soft tissue or Alcohol wipe
- Pen, vital sign flow sheet or record, or patient's electronic medical record
- Clean gloves, plastic thermometer sleeve, disposable probe or sensor cover
- Towel.

N	STEPS	RATIONALE
1.	Identify the patient.	To give care to the correct patient.
2.	Assess the site for temperature.	Help in identifying the most appropriate site for reading temperature.
3.	Wash hands.	Reduces risk of cross infection.
4.	Collect equipment.	Saves time and energy.

5.	Assist client into a comfortable position, explain procedure, and gain consent	Relieves patient's fears and anxiety moreover it gains cooperation.

A. Oral method



6.	Ensure patient has not taken hot / cold fluids and not smoked for at least 10 – 15 minutes.	Ensures correct reading.
7.	Hold mercury thermometer at eye level, rotating slightly to ensure mercury line is visible. Check mercury is low enough to record the temperature. If not, shake it down in a downward direction (35°C)	For accuracy of measurement.
8.	Place thermometer under the client's tongue beside the frenulum.	To ensure correct reading.
9.	Instruct the patient to: Close his / her mouth carefully with lips held firmly together. Avoid biting down the thermometer. Refrain from speaking.	To ensures accurate results and prevents thermometer from falling out and breaking down.
10.	Leave the thermometer in place for 3 – 5 minutes.	To ensure correct reading.
11.	Grasp the stem of the thermometer, ask the patient to open his /her mouth, remove the thermometer.	Less likely to chip on the patients' teeth or break the thermometer.
12.	Wipe off any secretion from thermometer with cotton swab / tissue. Wipe in rotating fashion from finger to bulb end.	Wiping allows clear reading of thermometer. Wiping is done from area of least contamination to area of greater contamination.

13.	Read the thermometer at eye level (by slowly rotating).	Ensure accurate reading.
14.	Clean thermometer by alcohol swab from clean to unclean part/ or according to local policy.	To minimize cross-infection
15.	Record result noting any significant change and report accordingly.	To ensure continuity of care and prompt attention if necessary

B. Axillary method

6	Put the curtains around patient's bed or close door (as required)	Provides privacy and comfort.
7	Ensure that axilla is dry.	Moisture conducts heat, and may give an inaccurate reading.
8	Move clothing or gown away from patient's shoulder and arm.	Provides optimal exposure of axilla.
9	Place the thermometer in center of axilla, lower the patient's arm over the thermometer, and place the forearm across the chest.	Maintains proper position of thermometer against blood vessels.
10	Gently hold the arm in place (if required).	Movement can displace thermometer can give false reading and thermometer can fall and break.
11	Leave the thermometer in place for a minimum of 5 – 10 minutes.	Ensure accurate reading.
12	Remove the thermometer, raise it to eye level, and note the reading.	Ensure accurate reading.
13	Wash thermometer with soap and warm water using firm twisted motion. Rinse with cold water/ or according to local policy.	Avoid contact of microorganisms with nurse's hand.

	Dry it with cotton swab / tissue using firm twisted motion.	
15	Inform client of temperature reading.	Promotes participation in care and understanding of health status.
16.	Replace thermometer in the provided case / antiseptic solution.	Storage container prevents breakage.
17.	Wash hands.	Reduce transmission of microorganisms
18.	Document accurately on flow sheet.	To ensure continuity of care and prompt attention if necessary

Checklist oral/ axillary body temperature

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2. Assess the site for temperature.					
3. Wash hands.					
4. Collect equipment.					
5.Assist client into a comfortable position, explain procedure, and gain consent					
A. Oral method					
6. Ensure patient has not taken hot / cold fluids and not smoked for at least 10 – 15 minutes.					
7. Check mercury is low enough to record the temperature (35°C).					
8. Place thermometer under the client's tongue beside the frenulum.					
9. Instruct the patient to: Close his / her mouth. Avoid biting down the thermometer.					
10. Leave the thermometer in place for 3 – 5 minutes.					
11. Grasp the stem of the thermometer, asks the patient to open his /her mouth, remove the thermometer.					
12. Wipe off any secretion from thermometer with cotton swab / tissue. Wipe in rotating fashion from finger to bulb end.					

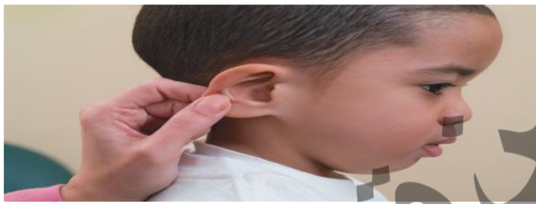
13. Read the thermometer at eye level (by slowly rotating).					
14. Clean thermometer by alcohol swab from clean to unclean part/ or according to local policy.					
15. Record result noting any significant change and report accordingly.					
B. Axillary method					
6. Close door.					
7. Ensure that axilla is dry.					
8. Move clothing or gown away from patient's shoulder and arm.					
9. Place the thermometer in center of axilla, lower the patient's arm over the thermometer, and place the forearm across the chest.					
10. Gently hold the arm in place.					
11. Leave the thermometer in place for a minimum of 5 – 10 minutes.					
12. Remove the thermometer, raise it to eye level, and note the reading.					
13. Wash thermometer with soap and warm water. Rinse with cold water/ or according to local policy.					
14. Dry it with cotton swab / tissue using firm twisted motion.					
15. Replace thermometer in the case.					
16. Wash hands.					
17. Document accurately on flow sheet.					

C. Tympanic membrane temperature with electronic infrared thermometer

Equipment:

- Tympanic membrane thermometer.
- Disposable protective probe cover.
- Paper and pen; patient’s medical record.
- Waste container.
- Clean gloves.
- Disposable probe or sensor cover.
- Alcohol swab.

Steps	Rational
1. Identify the patient.	To give care to correct patient.
2. Wash hands.	Reduces risk of cross infection.
3. Collect equipment.	Saves time and energy.
4 .Help patient assume comfortable position with head turned toward side away from you. If patient has been lying on side, use upper ear.	Ensures comfort and exposes auditory canal for accurate temperature measurement.
5. Note if there is obvious earwax in patient’s ear canal.	Earwax on lens cover blocks a clear optical pathway.
6. Remove the thermometer from its base. The display should read “Ready.”	Prepares it to measure temperature
7. Attach a disposable probe cover to the earpiece. 	Soft plastic probe cover prevents transmission of microorganisms between patients.

8. With one hand, gently pull upward and out on the patient's outer ear if an adult. Pull back and downward if the patient is an infant or child.	Straightens the external auditory canal; allows maximum exposure of the tympanic membrane. 
9 .Gently insert the plastic-covered tip of the probe into the ear canal.	
10. Activate the thermometer by pressing the scan button. Observe the temperature reading in the display window.	Pressing scan button causes detection of infrared energy.
11. Gently withdraw the thermometer from the ear canal.	Prevents rubbing of sensitive outer ear lining.
12. Dispose of the used probe cover into a waste container by pressing the eject button.	Reduces transmission of microorganisms.
13. If temperature is abnormal a second reading is necessary. Repeat in another ear.	To accurate measure
14. Help patient get into a comfortable position.	Restores comfort and sense of well-being.
15. Return the tympanic membrane thermometer to its base.	Automatically causes digital reading to disappear and prevents damage to Sensor.
16. Perform hand hygiene.	Reduces transmission of microorganisms
17. Record the temperature in the patient's medical record, and which ear was used. -Report any abnormality finding.	To ensure continuity of care and prompt attention if necessary

Checklist tympanic membrane temperature with electronic infrared thermometer

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2. Wash hands.					
3. Collect equipment.					
4. Help the patient to head turned toward side away from you.					
5. Note if there is obvious earwax in patient's ear canal.					
6. Remove the thermometer from its base. The display should read "Ready."					
7. Attach a disposable probe cover to the earpiece.					
8. With one hand, gently pull upward and out on the patient's outer ear if an adult. Pull back and downward if the patient is an infant or child.					
9 .Gently insert the plastic-covered tip of the probe into the ear canal.					
10. Activate the thermometer by pressing the scan button.					
11. Gently withdraw the thermometer from the ear canal.					
12. Dispose of the used probe cover into a waste container.					
13. If temperature is abnormal. Repeat in another ear.					
14. Help patient get into a comfortable position.					
15. Return the tympanic membrane thermometer to its base.					
16. Perform hand hygiene.					
17. Document, and report any abnormality. - Documentation provides coordination of care					

D. Rectal temperature measurement with electronic thermometer

Equipment:

- Gloves.
- Disposable bag.
- Lubricant.
- Appropriate chart.
- Appropriate thermometer.
- Modesty sheet/towel to protect patient's dignity.

Steps	Rationale
1. Identify the patient.	To give care to correct patient.
2. Wash hands.	Reduces risk of cross infection.
3. Collect equipment.	Saves time and energy.
4. Screen the bed/close door.	Promotes comfort and dignity; minimizes embarrassment.
5. Assist client into lateral position with upper leg flexed; keep majority of client covered; expose anal area only.	For patient comfort and dignity.
6. Apply lubricant to a tissue and dip end of thermometer into lubricant.	Minimizes trauma to rectal mucosa during insertion. Using tissue avoids contamination of tube/container.
7. Ask client to relax and take deep breaths With non-dominant hand separate client's buttocks to expose anus.	Relaxes anal sphincter for ease of insertion.

8. Insert the thermometer no more than 5 cm into the rectum and hold the thermometer in place. Allow the thermometer time to register (minimum two minutes). Note the Reading	To prevent trauma. To allow adequate time for the thermometer to register. 
9. Wipe client's anal area with a soft tissue; remove gloves and dispose in waste bag.	To promote comfort and prevent cross-infection.
10. Assist the client into a comfortable position and record on measurement chart.	To promote client comfort and adhere to legislation surrounding record keeping.
11. Clean thermometer, adhering to local policy.	To minimize the risk of cross-infection.
12. Perform hand hygiene.	Reduces transmission of microorganisms.
13. Document, Report any significant change or abnormality	To ensure prompt attention.

Checklist Rectal temperature measurement with electronic thermometer

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2. Wash hands.					
3. Collect equipment.					
4. Screen the bed/close door.					
5. Assist client into lateral position with upper leg flexed; expose anal area only.					
6. Apply lubricant to a tissue and dip end of thermometer into lubricant.					
7. Ask client to relax and take deep breaths. With non-dominant hand separate client's buttocks to expose anus.					
8. Insert the thermometer no more than 5 cm into the rectum and hold the thermometer in place. Allow the thermometer time to register (minimum two minutes). Note the reading.					
9. Wipe client's anal area with a soft tissue; remove gloves and dispose in waste bag.					
10. Assist the client into a comfortable position.					
11. Clean thermometer, adhering to local policy.					
12. Perform hand hygiene.					
13. Document, Report any significant change or abnormality					

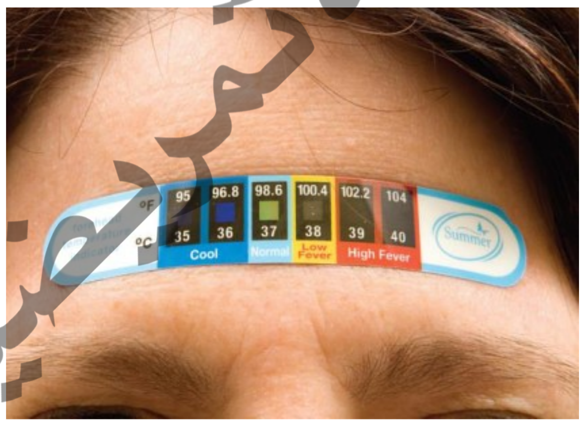
E. Temporal artery temperature measurement with electronic infrared thermometer

Steps	Rational
1. Identify the patient.	To give care to correct patient.
2. Wash hands.	Reduces risk of cross infection.
3. Collect equipment.	Saves time and energy.
4. Ensure that forehead is dry; wipe with towel if needed.	Moist skin interferes with thermometer sensor
5. Remove the cap from the probe of the thermometer, and disinfect the probe by gently wiping it with an alcohol swab.	To prevent infection.
6. Place sensor flush on patient’s forehead above eyebrow.	Contact avoids measurement of ambient temperature. 
7. Press red scan button with your thumb. Slowly slide thermometer straight across forehead while keeping sensor flush on skin.	Scanning for highest temperature continues until you release scan button.
8. Keeping scan button pressed, lift sensor from forehead and touch sensor to skin on neck, just behind earlobe. Peak temperature occurs when clicking sound during scanning stops. Release scan button and note reading.	Sensor confirms highest temperature behind earlobe.
9. Clean sensor with alcohol swab.	Prevents transmission of microorganisms.
10. Return thermometer to charger or thermometer base.	Maintains battery charge of thermometer unit.
11. Perform hand hygiene.	Reduces transmission of microorganism
12. Document, Report any abnormality	To ensure prompt attention.

Checklist Temporal artery temperature measurement with electronic infrared thermometer

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2. Wash hands.					
3. Collect equipment.					
4. Ensure that forehead is dry; wipe with towel if needed.					
5. Remove the cap from the probe of the thermometer, and disinfect the probe by gently wiping it with an alcohol swab.					
6. Place sensor flush on patient's forehead above eyebrow.					
7. Press red scan button with your thumb. Slowly slide thermometer straight across forehead while keeping sensor flush on skin.					
8. Keeping scan button pressed, lift sensor from forehead and touch sensor to skin on neck, just behind earlobe. Peak temperature occurs when clicking sound during scanning stops. Release scan button and note reading.					
9. Clean sensor with alcohol swab.					
10. Return thermometer to charger.					
11. Perform hand hygiene.					
12. Document, Report any significant change or abnormality					
13. Document, Report any significant change or abnormality					

F. Temperature Using a Heat-Sensitive Wearable Thermometer

Steps	Rational
1. Identify the patient.	To give care to correct patient.
2. Wash hands.	Reduces risk of cross infection.
3. Collect equipment.	Saves time and energy.
4. Ensure that forehead is dry; wipe with towel if needed.	Moist skin interferes with thermometer sensor
5. Place the thermometer strip on the forehead and begin timing for 15 seconds.	To measure. 
6. After 15 seconds, read the correct temperature by reading the color changes.	
7. Discard the strip in the waste container.	Prevents transmission of microorganisms.
8. Perform hand hygiene.	Reduces transmission of microorganism
9. Document, Report any abnormality	To ensure prompt attention.

Checklist Temperature Using a Heat-Sensitive Wearable Thermometer

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2. Wash hands.					
3. Collect equipment.					
4. Ensure that forehead is dry; wipe with towel if needed.					
5. Place the thermometer strip on the forehead and begin timing for 15 seconds.					
6. After 15 seconds, read the correct temperature by reading the color changes.					
7. Discard the strip in the waste container.					
8. Perform hand hygiene.					
9. Document, Report any abnormality					

2- Pulse measurement

Objectives:

Students will be able to:

- Define Pulse.
- List Purpose to pulse measurement.
- Describe characteristics of pulse rate.
- Identify factors that affect pulse rate.
- Identify the nine pulse site locations on the human body.
- Demonstrate measuring pulse procedure.

Definition:

Pulse is pressure of blood pushing against wall of artery as heart beats and rests per minute (bpm).

The normal pulse rate in adult is (60 – 100 beat/min.).

Purpose to pulse measurement:

1. To determine number of heart beats occurring per minute (rate).
2. To gather information about heart rhythm and pattern of beats.
3. To evaluate strength of pulse.
4. To assess heart's ability to deliver blood to distant areas of the blood viz. fingers and lower extremities.
5. To assess response of heart to cardiac medications, activity, blood volume and gas exchange.
6. To assess vascular status of limbs.

Characteristics of Pulse (Normal/ Abnormal):

- **1. Pulse quality:** refers to the ‘‘feel’’ of the pulse, its rhythm and forcefulness
- **2. Pulse rate:** is an indirect measurement of cardiac output obtained by counting the number of apical or peripheral pulse waves over a pulse point.

- **A normal pulse rate for adults** is between 60 and 100 beats per minute.
- **Bradycardia:** is a heart rate less than 60 beats per minute in an adult.
- **Tachycardia:** is a heart rate in excess of 100 beats per minute in an adult.

➢ **3. Pulse rhythm:** is the regularity of the heartbeat. It describes how evenly the heart is beating:

- **Regular** (the beats are evenly spaced).
- **Irregular** (the beats are not evenly spaced).
- **Dysrhythmia (arrhythmia):** is an irregular rhythm caused by an early, late, or missed heartbeat.

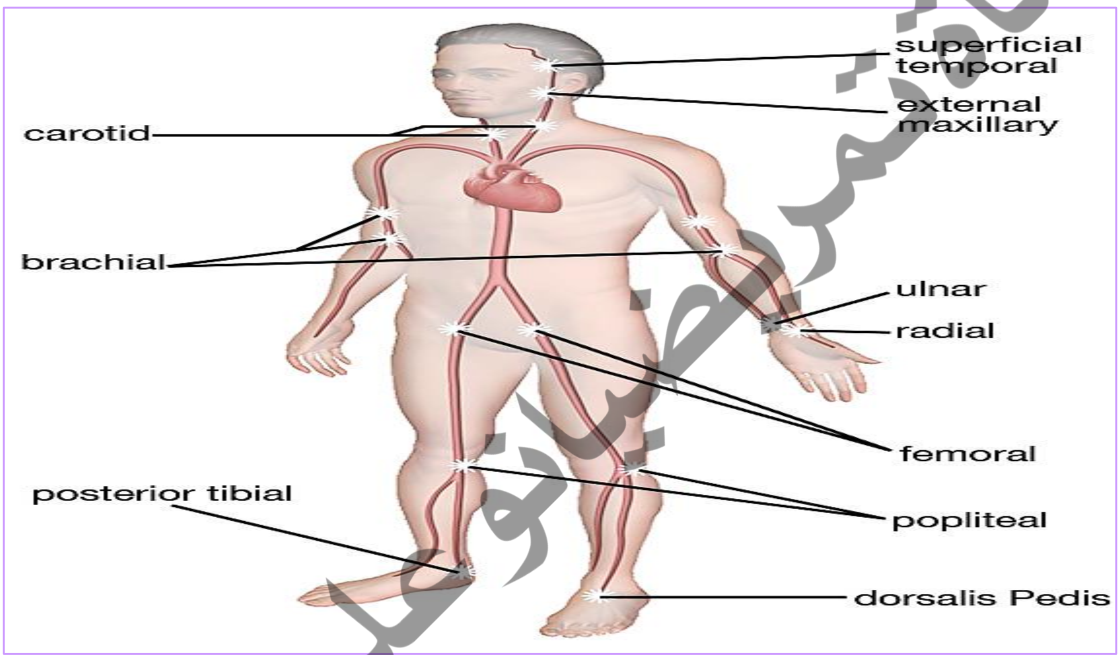
➢ **4. Pulse volume:** is a measurement of the strength or amplitude of force exerted by the ejected blood against the arterial wall with each contraction.

- **It is described as normal (full, easily palpable).**
- **Weak (49hread and usually rapid), or**
- **Strong (bounding).**

Factors That Influence Pulse Rate	
Exercise	Activity increases pulse rate. Rate may increase 20–30 bpm, based on the intensity of activity.
Age	As age increases, pulse rate decreases. Infants and children have a faster pulse rate than adults.
Gender	Female pulse rate is about 10 bpm higher than a male of the same age.
Physical condition	Athletes and people in good physical condition have lower pulse rates.
Size	Pulse rate is proportionate to the size of the body. Heat loss is greater in a small body, resulting in the heart pumping faster to compensate.
Disease condition	Pulse rate is increased in certain disease conditions such as thyroid disease, fever, and shock.

Medication	Many medications can either raise or lower the pulse rate. Medications such as digoxin are given to regulate the heartbeat. Caffeine and nicotine can increase the heart rate in certain people.
Depression	May lower the pulse rate.
Fear, Anxiety, Anger	May raise the pulse rate.

Sites of Taking Pulse Rate:



Location of Common Pulse Sites	
Radial	Radial Thumb side of wrist about 1 inch below base of thumb (most frequently used site).
Brachial	Brachial Inner (antecubital fossa/space) aspect of the elbow (pulse heard when taking BP).
Carotid.	Carotid At side of neck between larynx and sternocleidomastoid muscle (pulse used in CPR).
Temporal.	Temporal At side of head just above the ear
Femoral.	Femoral In groin where femoral artery passes to leg
Popliteal.	Popliteal Behind the knee; pulse located deeply behind the knee and felt when knee is slightly bent
Posterior Tibial.	Posterior Tibial On medial surface of ankle near ankle bone
Dorsalis Pedis.	Dorsalis Pedis On top of foot slightly lateral to midline; helps assess adequate blood circulation to the foot
Apical.	Apical At apex of heart; left of sternum, 4^{th} or 5^{th} intercostal space below the nipple

Measuring Radial / Apical pulse

Equipment:

- Stethoscope (apical pulse only).
- Wristwatch with second hand or digital display.
- Pen, vital sign flow sheet, or patient's electronic health record.

Radial pulse

N	STEPS	RATIONALE
1.	Identify the patient.	To give care to the correct patient.
2.	Wash hands.	Prevents the risk of cross infection.
3.	Prepare the equipment.	Saves time and energy and prevents interruption during procedure.

4.	Explain procedure to patient.	Gains cooperation, reduced patient's anxiety.
5.	Place patient in a comfortable position.	Relaxed position of lower arm and extension of wrist permits full exposure of artery for palpation.
6.	Place tips of first two or middle three fingers of dominant hand over groove along radial or thumb side of patient's inner wrist.	Fingers tip are most sensitive parts of hand to palpate arterial pulsation. Thumb has pulsation that may interfere with accuracy. 
7.	Lightly compress against radius; press pulse initially, and then relax pressure so pulse becomes easily palpable.	Pulse is more accurately assessed with moderate pressure. Too much pressure occludes pulse and impairs blood flow.
8.	Using a watch with a second hand, count the number of pulsations felt for 1 minute.	Sufficient time is necessary to assess the rate, rhythm and amplitude of the pulse.
9.	Assess the pulse, rhythm, and amplitude while counting rate.	Irregularity in heart rate may disrupt the cardiac output. Amplitude of pulse indicates the quality of the heart's contraction.
10.	Wash hands.	Prevents risk of cross infection.
11.	Record pulse **rate** on flow sheet, **site, rhythm** and **amplitude** in nurse's notes (if abnormal).	Promotes continuity of care.

Apical pulse

N	STEP	RATIONAL
1	Identify the patient.	To give care to the correct patient.
2	Wash hands.	Prevents the risk of cross infection.
3	Prepare the equipment.	Saves time and energy and prevents interruption during procedure.
4	Explain procedure to patient.	Gains cooperation, reduced patient's anxiety.
5	Clean earpieces and diaphragm of stethoscope with alcohol swab.	Reduces transmission of microorganisms.
6	Draw curtain around bed and/or close door	Maintains privacy
7	Help patient to supine or sitting position. Move aside bed linen and gown to expose sternum and left side of chest.	Provides easy access to pulse sites.
8	Place diaphragm of stethoscope in palm of hand for 5 to 10 seconds.	Warming of metal or plastic diaphragm prevents patient from being startled and promotes comfort
9	Place diaphragm of stethoscope over apical impulse at fifth intercostal space at left midclavicular line and auscultate for heart sounds.	Determine apical site accurately to be able to auscultate sounds clearly. 
10	If apical rate is regular, count for 30 seconds and multiply by 2.	Regular rate is accurate when measured for 30 seconds.
11	Note if heart rate is irregular and describe pattern or irregularity.	Irregular heart rate indicates dysrhythmia. Regular occurrence of dysrhythmia within 1 minute

12	Replace patient's gown and bed linen; help patient return to comfortable position.	Provides privacy and minimizes embarrassment.
13	Clean earpieces and diaphragm of stethoscope with alcohol swab routinely after each use.	Prevents transmission of microorganisms.
14	Perform hand hygiene.	Perform hand hygiene. Reduces transmission of microorganisms.
15	Record pulse **rate** on flow sheet, **site, rhythm** and **amplitude** in nurse's notes (if abnormal).	Promotes continuity of care.

Checklist radial pulse

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2. Wash hands.					
3. Prepare the equipment.					
4. Explain procedure to patient.					
5. Place patient in a comfortable position.					
6. Place tips of first two or middle three fingers of dominant hand over groove along radial or thumb side of patient’s inner wrist.					
7. Lightly compress against radius; press pulse initially, and then relax pressure.					
8. Using a watch with a second hand, count the number of pulsations felt for 1 minute.					
9. Assess the pulse, rhythm, and amplitude while counting rate.					
10. Wash hands.					
11. Record pulse **rate** on flow sheet, **site, rhythm** and **amplitude** in nurse’s notes (if abnormal).					

Checklist Apical pulse

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2. Wash hands.					
3. Prepare the equipment.					
4. Explain procedure to patient.					
5. Clean earpieces and diaphragm of stethoscope with alcohol swab.					
6. close door					
7. Help patient to supine or sitting position. Move aside bed linen and gown to expose sternum and left side of chest.					
8. Place diaphragm of stethoscope in palm of hand for 5 to 10 seconds.					
9. Place diaphragm of stethoscope over apical impulse at fifth intercostal space at left midclavicular line and auscultate for heart sounds.					
10. Note if heart rate is irregular and describe pattern or irregularity.					
11. Replace patient's gown and bed linen; help patient return to comfortable position.					
12. Clean earpieces and diaphragm of stethoscope with alcohol swab routinely after each use.					
13. Perform hand hygiene.					
14. Record pulse **rate**, **site, rhythm** and **amplitude** in nurse's notes (if abnormal).					

3- RESPIRATION

Objectives:

Students will be able to:

- Define respiration.
- List Purpose to respiration measurement.
- Describe characteristics of respiration rate.
- Identify factors that affect respiratory rate.
- Demonstrate measuring respiration procedure.

Definition:

Respiration, or the act of breathing, is the process of inhaling oxygen into the body and exhaling carbon dioxide.

Respiratory cycle: consists of one expiration (exhalation) and one inspiration (inhalation).

Purposes to respiration measurement:

1.To determines number of respirations occurring per minute.
2. To gather information about rhythm and depth.
3. To assess response of patient to any related therapy/ medication.

Characteristics of Respiration (Normal/ Abnormal):

- **Respiratory rate:** is the number of respirations per minute.

The normal respiration rate for healthy adults at rest is 12 to 20 cycles per minute.

Tachypnea: is a respiratory rate greater than 20 breaths per minute.
Bradypnea: is a respiratory rate of 12 or fewer breaths per minute.

- **Respiratory rhythm:** refers to the regular and equal spacing of breaths

Regular respiratory rhythm: the cycles of inspiration and expiration have about the same rate and depth.

Irregular breathing patterns: the depth and amount of air inhaled and exhaled and the rate of respirations per minute will vary.

- **Respiratory Depth:** The depth of respiration is the volume of air that is inhaled and exhaled. It is described as either "shallow" or "deep."

  Hyperventilation: is characterized by deep, rapid respirations.

  Hypoventilation: is characterized by shallow respirations.

- **Respiratory Quality:** refers to breathing patterns— both normal and abnormal.

  Labored breathing refers to respirations that require greater effort from the patient.

Factors affecting Respiratory Rate:

- **Increased Rate:**
  - Allergic reactions.
  - Disease (e.g., asthma, heart disease).
  - Excitement/anger.
  - Hemorrhage.
  - Obstruction of air passage.
  - Shock.
  - Certain drugs (e.g., epinephrine).
  - Exercise.
  - Fever.
  - Nervousness.
  - Pain.
- **Decreased Rate:**
  - Certain drugs (e.g., morphine).
  - Decrease of CO2 in blood.
  - Disease (stroke, coma).

Measuring respiratory rate

Equipment:

- Wristwatch with second hand or digital display
- Pen, vital sign flow sheet or record, or electronic health record

N	STEPS	RATIONALE
1.	Identify the patient.	To give care to the correct patient.
2.	Wash hands	Prevents the risk of cross infection.
3.	Collect equipment.	Saves time and energy
4.	Be sure client's chest is visible remove bed linen or gown.	
5.	Assess patient's activity prior to checking respiration.	A patient who has been exercising will needed to rest for few minutes to permit the accelerated respiratory rate return to normal.
6.	Place patient in a comfortable position.	
7.	Place hand against patient's chest to feel his chest movement or place patient's arm across the chest and observe the chest movement while supposedly taking the radial pulse.	Awareness of respiratory rate assessment would cause the patient voluntarily to alter the respiratory pattern. 
8.	Check the respiratory rate, rhythm and depth for 1 minute.	Accuracy of reading.
9.	Wash hands.	Prevents the risk of cross infection.
10.	Document in flow sheet and (if required) nurse's note, report any abnormality.	Promotes continuity of care.

Checklist respiratory rate measurement

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2.Wash hands					
3. Collect equipment.					
4. Be sure client's chest is visible remove bed linen or gown.					
5. Assess patient's activity prior to checking respiration.					
6. Place patient in a comfortable position.					
7. Place hand against patient's chest to feel his chest movement or place patient's arm across the chest and observe the chest movement while supposedly taking the radial pulse.					
8. Check the respiratory rate, rhythm and depth for 1 minute.					
9. Perform hand hygiene.					
10. Record respiration **rate**, **rhythm** and **report** if any abnormality.					

Blood pressure

Objectives:

Students will be able to:

- Define blood pressure.
- List purposes of measuring blood pressure.
- Identify factors that affect blood pressure.
- Differentiate between types of sphygmomanometers
- Explain phases of Korotkoff Sounds.
- Demonstrate measuring blood pressure procedure.

Definitions:

Blood pressure is the force required by the heart to pump blood from the ventricles of the heart into the arteries. It is measured in systolic and diastolic pressure.

Systolic pressure: is the highest pressure that occurs as the left ventricle of the heart is contracting.

Diastolic blood pressure: is the lowest pressure level that occurs when the heart is relaxed and the ventricle is at rest and refilling with blood.

- The pulse beat is felt (or heard) at the systolic pressure level and is absent at the diastolic pressure level.
- Blood pressure is read in millimeters (mm) of mercury (Hg), or “mmHg,”

Pulse pressure: is the difference between the systolic and diastolic readings and calculated by subtracting the diastolic reading from the systolic reading. If the blood pressure is 120/80, the pulse pressure is 40.

- In general, a pulse pressure that is **greater than 40 mmHg** is considered widened, and one that is **less than 30 mmHg** is considered to be narrowed.

Purpose to measure blood pressure:

- To maintain a base line measure of arterial pressure.
- To assess the hemodynamic (i.e., the study of movement of blood and the forces concerned) status of a patient.
- To monitor response of the circulatory system to various disease conditions and therapies.

Blood Pressure Guidelines:

Ideally, blood pressure is measured while the patient is seated upright with both feet flat on the floor. Sitting with legs crossed at the knees can elevate blood pressure readings. If warranted by the patient’s condition, blood pressure may be measured while lying on an examination table. A rise or fall in blood pressure is an indication of many medical conditions.

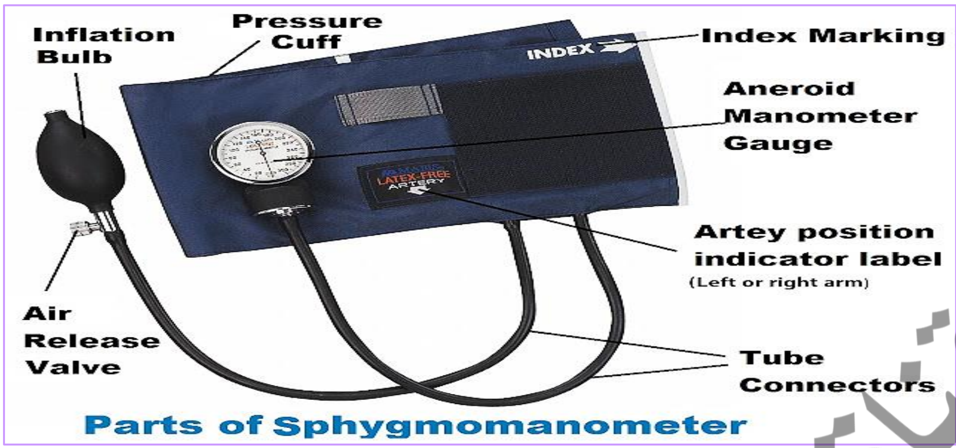
Blood Pressure Categories

BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)		DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130 – 139	or	80 – 89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
HYPERTENSIVE CRISIS (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

Factors affecting blood pressure:

1. **Age:** in older adults, the diastolic pressure often increases as a result of the reduced compliance of the arteries.
2. **Exercise:** Physical activities increase both the cardiac output and hence blood pressure, thus, a rest of 20 to 30 minutes following exercise is indicated before the blood pressure can be reliably assessed.
3. **Stress:** thus, increasing the blood pressure reading.
4. **Severe pain**: can decrease blood pressure greatly and cause shock.
5. **Obesity.**
6. **Sex:** after puberty, females usually have lower blood pressure than males of the same age. After menopause, women generally have higher blood pressure than before.
7. **Medications:** many medications may increase or decrease the blood pressure.
8. **Diseases** (Fever, hemorrhage…. etc.).

Types of sphygmomanometers:



Korotkoff Sounds:

Korotkoff sounds are the rhythmic, tapping sounds heard while taking blood pressure as the arterial wall distends under the compression of the cuff.

Phases (Korotkoff's Sounds Correlated to Pressure Dynamics):

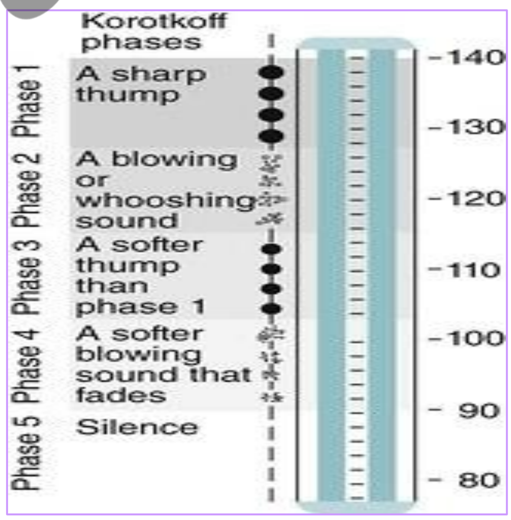
Phase I: The period initiated by the first faint clear taping sound. These sound gradually become more intense.

Phase II: The period during which the sounds have a swishing quality.

Phase III: The period during which the sounds are crisper and more intense.

Phase IV: The period, during which the sounds become muffled and have a soft, blowing quality.

Phase V: The period where the muffled, blowing sound disappear.



Measuring blood pressure

Equipment:

- Sphygmomanometer.
- Cloth or disposable vinyl pressure cuff of appropriate size for patient's extremity.
- Stethoscope.
- Alcohol swab.
- Pen, vital sign flow sheet or record form, or electronic medical record.

N	STEPS	RATIONALE
1.	Identify the patient.	To give care to the correct patient.
2.	Explain procedure to patient.	Reduces anxiety and gains cooperation.
3.	Collect and check equipment.	Ensures proper functioning of apparatus.
4.	Have patient in sitting or supine position.	Promotes comfort and relaxes patient. Provides accurate reading.
5.	Wash hands.	Prevents cross infection.
6.	Clean the ear piece of stethoscope with the spirit swab.	Prevents transmission of microorganisms. 
7.	Be sure that the manometer is positioned vertically at eye level.	Ensure accurate reading of mercury level.
8.	Support patients fore-arm at heart level, with palm turned up.	Blood pressure increases when the arm is below heart level and decreases when the arm is above heart level.

9.	Expose patient's left upper arm by removing constricting clothing.	Ensures proper cuff application. Tight sleeves interfere with the ability to hear pulsations and may cause inaccurate readings.
10.	Warp the deflated cuff evenly around the upper arm by placing the lower edge of the cuff 2.5 –5 cm above the antecubital space. 	Even wrapping produces equal pressure. Too loose / tight cuff will give inaccurate reading. The bladder directly over the brachial artery gives accurate reading.
11.	Place the stethoscope in your ear and check the diaphragm by tapping.	Tapping is done to check whether the sound is audible. 
12.	Make sure to unlock the mercury column before inflating.	
13.	Palpate brachial or radial pulse with one hand. Close the valve of the bulb: inflate the cuff noting the level of mercury where pulse disappears.	Indicates approximate systolic pressure (Done if it is the initial examination).
14.	Deflate cuff quickly and wait for 30 seconds; tighten the valve.	Prevents venous congestion and false high reading.
15.	Relocate brachial artery and place the diaphragm of stethoscope over the	Proper stethoscope placement ensures optimal sound reception. Improper position

	brachial pulse and hold it in place (Do not let the diaphragm touch the cuff or patient's clothing).	of diaphragm causes muffled sounds and often results in false low systolic and false high diastolic readings. 
16.	Inflate the cuff to 30 mm Hg above where the pulse disappeared.	Ensures accurate measurement of systolic pressure.
17.	Slowly release the valve and allow mercury to fall at the rate of 2 – 3 mm Hg per sec.	Too rapid or too slow decline of mercury level can cause inaccurate readings.
18.	Note point on manometer when first clear sound is heard.	First Korotkoff sound indicates the systolic pressure.
19.	Continue to deflate cuff gradually noting point at which sounds disappear.	This is noted as the diastolic pressure.
20.	Deflate cuff rapidly and completely. Remove cuff from patient's arm. (Lock mercury column unless measurement must be repeated.)	Prevents arterial occlusion resulting in numbness and tingling of patient's arm.
21.	Assist patient to a comfortable position. Cover the upper arm.	Ensure patients comfort
22.	Inform client of B.P reading (depends on patient's conditions).	Promotes participation in care and understanding of health status.
23.	Clean stethoscope with spirit swab and return equipment to appropriate place.	Prevents spread of microorganisms and safety of equipment.
24.	Wash hands.	Prevents transmission of microorganisms.
25.	Record accurately in the flow sheet according to hospital policy.	Timely documentation ensures accurate therapeutic intervention, if needed.

Checklist measuring blood pressure

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Identify the patient.					
2. Explain procedure to patient.					
3. Collect and check equipment.					
4. Have patient in sitting or supine position.					
5. Wash hands.					
6. Clean the ear piece of stethoscope with the spirit swab.					
7. Be sure that the manometer is positioned vertically at eye level.					
8.Support patients fore-arm at heart level, with palm turned up.					
9. Expose patient's left upper arm by removing constricting clothing.					
10. Warp the deflated cuff evenly around the upper arm by placing the lower edge of the cuff 2.5 –5 cm above the antecubital space.					
11. Place the stethoscope in your ear and check the diaphragm by tapping.					
12. Make sure to unlock the mercury column before inflating.					
13. Palpate brachial or radial pulse with one hand. Close the valve of the bulb: inflate the cuff noting the level of mercury where pulse disappears.					
14. Deflate cuff quickly and wait for 30 seconds; tighten the valve.					

15. Relocate brachial artery and place the diaphragm of stethoscope over the brachial pulse and hold it in place.					
16. Inflate the cuff to 30 mm Hg above where the pulse disappeared.					
17.Slowly release the valve and allow mercury to fall at the rate of 2 – 3 mm Hg per sec.					
18. Note point on manometer when first clear sound is heard.					
19. Continue to deflate cuff gradually noting point at which sounds disappear.					
20. Deflate cuff rapidly and completely. Remove cuff from patient’s arm.					
21. Assist patient to a comfortable position. Cover the upper arm.					
22. Clean stethoscope with spirit swab and return equipment to appropriate place.					
23. Wash hands.					
24. Record accurately in the flow sheet according to hospital policy.					

Pain assessment

Objectives

Student will be able to

- Define pain and Pain assessment.
- Identify types of pain.
- Enumerate factors contributing to pain and discomfort.
- Discuss effects of pain on patients.
- Demonstrate different methods of pain assessment.

➢ Definition of Terms

Pain is an unpleasant sensory and emotional experience, associated with, or resembling that associated with, actual or potential tissue damage

Pain assessment: is a multidimensional observational assessment of a patients' experience of pain.

Pain measurement tools: are instruments designed to measure pain.

➢ Factors Contributing to Pain and Discomfort

❖ Physical Factors

- Illness and injuries
- Wounds
- Sleep disturbance and deprivation
- Immobility
- Fever or hypothermia

❖ Psychosocial Factors

- Anxiety and depression
- Loss of control
- Impaired communication
- Fear of pain
- Fear of death
- Separation from family
- Unfamiliar surroundings

❖ Environmental Factors and routine

- Continuous noise
- Continuous light
- Awakening and physical manipulation every 1 to 2 hours for vital signs or positioning
- Continuous and frequent invasive procedures
- Competing priorities in care: unstable vital signs, bleeding, dysrhythmias and poor ventilation may take precedence over pain management.

➢ Types of Pain

We categorize pain in several ways. Most often, pain is categorized as nociceptive or neuropathic based on underlying pathology. Another useful scheme is to classify pain as acute or chronic

❖ Pain Categorized by Origin (underlying pathology)

- **Nociceptive** is the response of our bodies (sensory nervous systems) towards actual or potentially harmful stimuli.
- **Neuropathic pain** is described as a nerve injury or nerve impairment that causes a pain response from a stimulus that is non-painful in normal conditions.

❖ Pain Categorized by Nature (cause, course, manifestations, and treatment)

- **Acute pain** is defined as pain with sharp well-defined onset and short duration. It is usually caused by nociceptive or inflammatory cause.
- **Chronic pain** is defined as pain that prolongs from three to six months after the onset or the expected period of healing.

	Acute	**Chronic**
Time span	Less than 6 months	More than 6 months
Location	Localized, associated with a specific injury, condition, or disease	Difficult to pinpoint
Characteristics	Often described as sharp, diminishes as healing occurs	Often described as dull, diffuse, and aching
Physiological signs	• Elevated heart rate • Elevated BP • Elevated respirations • May be diaphoretic • Dilated pupils	• Normal vital signs • Normal pupils • No diaphoresis • May have loss of weight
Behavioral signs	• Crying and moaning • Rubbing site • Guarding • Frowning • Grimacing • Statement of pain	• Physical immobility • Hopelessness • Listlessness • Loss of libido • Exhaustion and fatigue • Expresses pain only when asked

➢ Consequences of Pain Patients

Unrelieved pain is the most traumatic memory patients. Under treatment of pain leads to harmful physiological and psychological effects.

❖ Physiological

- Tachypnea.
- Tachycardia.
- increased oxygen demand
- Ischemic injury.
- Delayed wound healing.
- Infections.
- Hyperglycemia.

❖ Psychological

- Anxiety
- Depression

❖ **Ethical**

- Breakdown of trust
- Suffering
- Failure to adhere to responsibility

❖ **Financial**

- Increased ventilated days
- Increased length of stay
- Increased total use of medication
- Increase chance of readmission

Body System	Effect
Cardiovascular	Tachycardia, Hypertension, Increase cardiac work load
Pulmonary	Respiratory muscle spasm, Decrease ventilation capacity, Atelectasis, Hypoxia, Increased risk of pulmonary infection
Gastrointestinal	Postoperative Ileus
Renal	Increased risk of oliguria and urinary retention
Coagulation	Increased risk of thromboembolism
Immunologic	Impaired immune function
Muscular	Muscle weakness, Fatigue, Limited mobility, Increase the risk of thromboembolism
Psychological	Anxiety, Fear, Frustration, Poor patient satisfaction

➢ Patient Barriers to Pain Assessment

- Communication
- Altered Level of Consciousness
- Older Patients
- Cultural Influences
- Lack of Knowledge

➢ Methods of pain assessment

Pain assessment is an integral part of nursing care. It is a prerequisite for adequate pain control and relief. Pain assessment has two major components:

1) subjective non-observable.

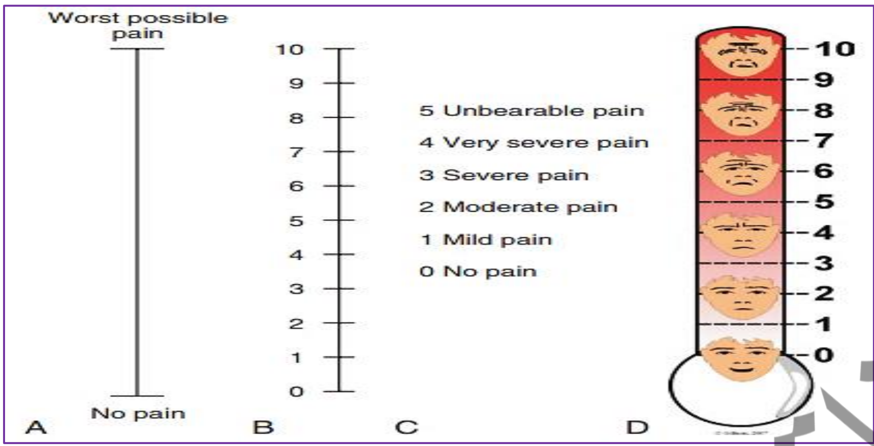
2) **objective or observable or.**

A. The Subjective Component

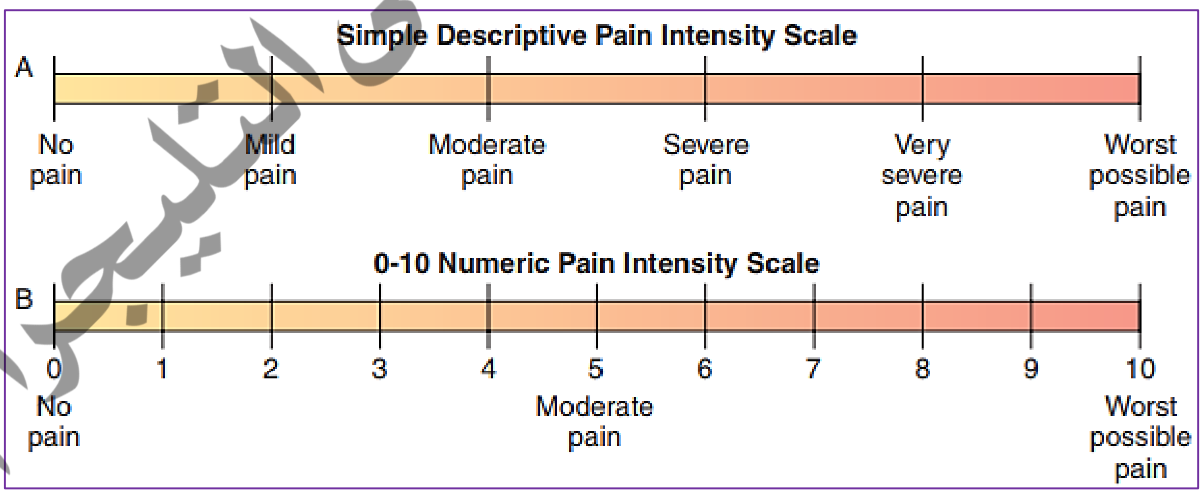
Pain is known as a subjective experience. The subjective component of pain assessment refers to the patient's self-report about his or her sensorial, affective, and cognitive experience of pain.

The patient's self-report of pain can also be obtained by questioning the patient using the mnemonic **PQRSTU**

- **Provocation Provocative and Palliative or Aggravating Factors**: indicates what provokes or causes the patient's pain what he or she was doing when the pain appeared, and what makes the pain worse or better
- **Quality**: characteristics of pain, refers to the quality of the pain or the pain sensation that the patient is experiencing. For instance, the patient may describe the pain as dull, aching, sharp, burning, or stabbing.
- **Region/ Radiation**: Where is the pain located? Does the pain radiate? Where? Does it feel like it travels/moves around? Did it start elsewhere and is now localized to one spot?
- **Severity:** How severe is the pain on a scale of 0 to 10, Does it interfere with activities? How bad is it at its worst?
- **Timing:** When/at what time did the pain start? How long did it last? How often does it occur: hourly? Daily? Weekly? Monthly? Is it sudden or gradual?
- **Understanding:** is the patient's perception of the problem or cognitive experience of pain.

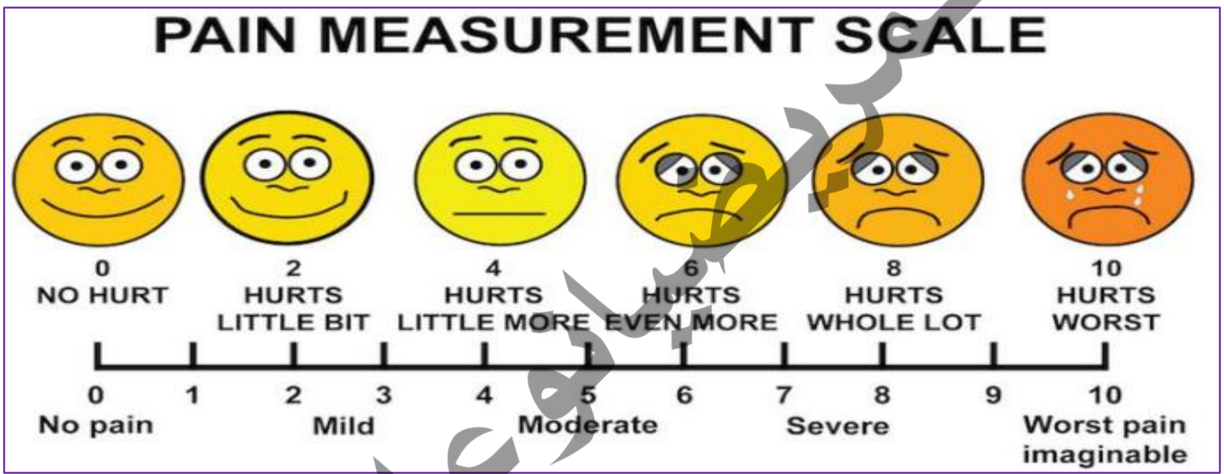


- **<u>Verbal Rating Scale (VRS)</u>**
    - Use words to describe pain. Word such as no pain, mild pain, moderate pain & severe pain are used to describe pain levels.
- **<u>Numeric Rating Scale (NRS)</u>**
    - A numerical rating scale with the range of 0 to 10 is another type of pain scale that is used.
    - The word "no pain" appear by "0" and "worst pain possible" is found by "10"
    - Patients are asked to choose a number from 0 to 10 that best reflects his/her level of pain.



➢ **Wong-Baker Faces Pain Rating Scale** with the Wong-baker pain scale, six faces are used:

- Face 0 is a happy face
- Face 2 is still smiling
- Face 4 is not smiling
- Face 6 is starting to frown
- Face 8 is definitely frowning
- Face 10 is crying



How to use?

Explain to the person that each face is for a person who feels happy because he has no pain (hurt) or sad because he has some or a lot of pain. Face 0 is very happy because he doesn't hurt at all. Face 2 hurts just a little bit. Face 4 hurts a little more. Face 6 hurts even more. Face 8 hurts a whole lot. Face 10 hurts as much as you can imagine, although you don't have to be crying to feel this bad. Ask the person to choose the face that best describes how he is feeling.

B. The Observable or Objective Component

When the patient's self-report is impossible to obtain, nurses can rely on the observation of behavioral indicators, which are strongly emphasized in clinical recommendations and guidelines for pain management in nonverbal patients.

- **Behavioural Pain Scale (BPS)**
  - The BPS is composed of 3 subscales: facial expression, movement of the upper limbs, and compliance with mechanical ventilation (MV).
  - Each subscale is scored from 1 (no response) to 4 (full response).
  - A BPS score of 5 or higher is considered to reflect unacceptable pain.

Behavioral Pain Scale (BPS)		
ITEM	**DESCRIPTION**	**SCORE**
Facial expression	Relaxed Partially tightened (e.g., brow lowering) Fully tightened (e.g., eyelid closing) Grimacing	**1** **2** **3** **4**
Upper limbs	No movement Partially bent Fully bent with finger flexion Permanently retracted	**1** **2** **3** **4**
Compliance with ventilation	Tolerating movement Coughing but tolerating ventilation for most of the time Fighting ventilator Unable to control ventilation	**1** **2** **3** **4**
Total		**3 to 12**
From Payen JF, et al. Assessing pain in the critically ill sedated patients by using a behavioral pain scale. Crit Care Med 29:(12) 2258-2263, 2001.		

➢ Critical-Care Pain Observation Tool (CPOT)

- It is the most valid and reliable for use in ICU patients who are unable to self-report.
- The CPOT has 4 components: facial expression, body movements, muscle tension, and compliance with the ventilator for intubated patients or vocalization for extubated patients.
- Each component is scored from 0 to 2 with a possible total score ranging from 0 to 8. A CPOT ≥ 3 is indicative of significant pain.

Indicator	Description	Score	
Facial expression	No muscular tension observed	Relaxed, neutral	0
	Presence of frowning, brow lowering, orbit tightening, and levator contraction	Tense	1
	All of the above facial movements plus eyelid tightly closed	Grimacing	2
Body movements	Does not move at all (does not necessarily mean absence of pain)	Absence of movements	0
	Slow, cautious movements touching or rubbing the pain site, seeking attention through movements	Protection	1
	Pulling tube, attempting to sit up, moving limbs/thrashing, not following commands, striking at staff, trying to climb out of bed	Restlessness	2
Muscle tension	No resistance to passive movements	Relaxed	0
Evaluation by passive flexion and extension of upper extremities	Resistance to passive movements	Tense, rigid	1
	Strong resistance to passive movements, inability to complete them	Very tense or rigid	2
Compliance with the ventilator (intubated patients)	Alarms not activated, easy ventilation	Tolerating ventilator or movement	0
OR	Alarms stop spontaneously	Coughing but tolerating	1
	Asynchrony: blocking ventilation, alarms frequently activated	Fighting ventilator	2
Vocalization (extubated patients)	Talking in normal tone or no sound	Talking in normal tone or no sound	0
	Sighing, moaning	Sighing, moaning	1
	Crying out, sobbing	Crying out, sobbing	2
Total, range			0–8

Source: American Association of Critical-Care Nurses (Gelinas, Fillion, Puntillo, Viens, & Fortier, 2006)

- **<u>FLACC - The acronym FLACC stands for Face, Legs, Activity, Cry and CONSOL ability .</u>**

How to use FLACC

Each category (Face, Legs etc) is scored on a 0-2 scale, which results in a total pain score between 0 and 10. The person assessing the child should observe them briefly and then score each category according to the description supplied.

FLACC has a high degree of usefulness for cognitively impaired and many critically ill children

	0	1	2
Face الوجه	No Particular Expression or Smile لا يوجد اي تغيير علي وجه المريض أو مبتسم	Occasional Grimace Withdrawn Disinterested. متجهم مكشر غير مهتم	Frequent to Constant Frown, Clenched Jaw. عبوس و يعض على فكية
Legs الأرجل	Normal Position or Relaxed وضع طبيعي أو مسترخي	Uneasy, Restless, Tense تتحرك بصعوبة و غير مرتاح و مشدود	Kicking Or Legs Drawn Up. يركل أو يضم أرجله علي صدره
Activity النشاط	Lying Quietly, Normal Position Moves Easily. ويتحرك بسهولة	Squirming Shifting Back & Forth, Tense.يكون في وضع القرفصاء و مشدود	Arched, Rigid, Or Jerking. مقوس,متيبس ويهتز
Cry البكاء	No Cry Awake Or asleep لا يبكي صاح أو نائم	Moans Or Whimpers Occasional Complaint. يئن أو يبكي و ممكن أن يشتكي	Crying Steadily, Screams, Frequent Complaints. الصراخ مع الشكوى ودائم البكاء
Consolability متماسك	Content, Relaxed. مقتنع أو مرتخي	Reassured By Occasional Touching Hugging To Talking To يطمئن بالكلام أو اللمس	Difficult To Console Or Comfort. صعوبة في الرضاء

Key considerations

- assess pain using a developmentally and cognitively appropriate pain tool
- reassess pain after interventions given to reduce pain (eg. Analgesia) have had time to work
- assess pain at rest and on movement
- investigate higher pain scores from expectation
- document pain scores
- use parent/guardian pain behavior knowledge for children with cognitive impairment

Oxygen therapy

Objectives

Student will be able to

- Define Oxygen therapy.
- List the Oxygen therapy devices.
- Enumerate Indications for Oxygen therapy.
- Demonstrate Oxygen therapy administration procedure for each method

- **Oxygen therapy** is the administration of oxygen, in concentrations greater than those in room air to treat hypoxia. These supplementary oxygen concentrations can vary from 24% (nasal cannula) to 95% (non-rebreathe oxygen mask)

- It is important to select the correct oxygen delivery device for the patient, i.e., how much supplementary oxygen needs to be administered and what the patient can actually tolerate

- **Oxygen** is a colorless, odorless gas that forms about 21% of the earth's atmosphere and is essential for plant and human life

- **Tissue oxygenation** is dependent upon inspired oxygen, the concentration of hemoglobin, and its ability to saturate with oxygen, as well as the circulation of blood

Indications

- Respiratory compromise
- Anaphylaxis
- Shock
- During anesthesia
- Post surgery

Equipment of oxygen therapy

- Oxygen supply: e.g., piped oxygen behind the bed, oxygen cylinder
- Flowmeter: to determine oxygen flow rate in liters/minute
- Oxygen tubing
- Oxygen delivery mechanism: e.g., nasal cannula, face masks, non-rebreathe oxygen mask
- Humidifier: to warm and moisten the oxygen prior to administration.
- Face shield as needed for risk of splash
- Clean gloves, if secretions are present
- Pulse oximeter

The oxygen delivery method selected depends on:

- Age of the patient
- Oxygen requirements/therapeutic goals
- Patient tolerance to selected interface
- Humidification requirement.

Procedure

Steps	Rational
1. Identify patient using at least two identifiers (e.g., name and birthday or name and medical record number) according to agency policy.	Ensures correct patient. Complies with The Joint Commission standards and improves patient safety
2. Perform hand hygiene.	Reduces transmission of microorganisms
3. Verify the physician's or qualified practitioner's order	Ensures correct dosage and route.
4. Explain procedure and hazards to the client.	Increases compliance with procedures.
5. If available, note patient's most recent arterial blood gas (ABG) results or pulse oximetry (SpO2) value.	Objectively documents the patient's pH, arterial oxygen and arterial carbon dioxide concentrations, or arterial oxygen saturation.

6. Attach oxygen delivery device (e.g., cannula, mask, T tube, tracheostomy collar) to oxygen tubing, and attach end of tubing to humidified oxygen source adjusted to prescribed flow rate. 	Humidity prevents drying of nasal and oral mucous membranes and airway secretions.
7. Insert humidifier and flow meter into oxygen source in wall or portable unit 	For access to oxygen. Many institutions also have compressed air available from outlets very similar in appearance to oxygen outlets. Green always stands for oxygen. Be sure to plug the flow meter into the green outlet.
8. Apply oxygen device: Attach the oxygen tubing and nasal cannula to the flow meter and turn it on to the prescribed flow rate (1–5 liters/min). Use extension tubing for ambulatory clients so they can get up to go to the bathroom.	Rates above 6 liters/min are not efficacious and can dry the nasal mucosa.

<table>
<tr>
<td>Places the Nasal cannula: in patient’s nares, then places the tubing around each ear. Uses the slide adjustment device to tighten the cannula under patient’s chin.
</td>
<td>Keeps delivery system in place so client receives the amount of oxygen ordered</td>
</tr>
<tr>
<td>Apply Simple face mask by placing it over patient’s mouth and nose. Then bring the straps over patient’s head and adjust to form a comfortable but tight seal.
</td>
<td>A properly fitting device that does not create pressure on nares or ears is comfortable,
and patient is more likely to keep it in place; reduces risk for skin breakdown</td>
</tr>
<tr>
<td>Partial or nonrebreather mask: Mask seals tightly around mouth. Reservoir fills on exhalation and almost collapses on inspiration. Reservoir should not collapse completely.
</td>
<td>Easily humidifies oxygen and does not dry mucous membranes. Useful for short-term therapy of 24 hours or less.</td>
</tr>
</table>

Venturi mask: Apply as regular mask. Select appropriate flow rate. 	It is used when high-flow device is desired.
Face tent: Apply tent under patient's chin and over the mouth and nose. It will be loose, and a mist is always present. 	Excellent source of humidification; however, you cannot control oxygen concentrations, and patient who requires high oxygen cannot use this device.
High-flow nasal cannula: Fit as for nasal cannula. 	Used when high oxygen delivery is required.

9. Makes sure that the oxygen equipment is set up correctly and functioning properly before leaving patient's bedside.	Ensures that client receives proper dose.
10. Check cannula/mask every 8 hours. Keep humidification container filled at all times.	Ensures patency of cannula and oxygen flow. Oxygen is a dry gas; when it is administered via nasal cannula of 4 L/ min or more, you must add humidification so patient inhales humidified oxygen
11. Monitor airway patency, vital signs, oxygen saturation, and for signs and symptoms of hypoxia every 2 hours. Additionally, monitor breath sounds and tube position every 4 hours.	Detects response to or any untoward effects from therapy. Determines whether tube is in place.
12. Properly dispose of gloves (if used) and perform hand hygiene.	Reduces transmission of microorganisms

Procedure Checklist

Steps	Mark	Trial 1	Trial 2	Trial 3	comment
1. Identifies the patient according to agency policy using two identifiers; perform hand hygiene, safety, privacy, body mechanics, and documentation.					
2. Attaches the flow meter to the oxygen source.					
3. Assembles the oxygen equipment.					
4. Attaches the humidifier to the flow meter. Humidification is necessary only for flow rates of greater than 3 L/min.)					
5. Turns on the oxygen using the flow meter and adjusts it according to the prescribed flow rate.					
Variation: Nasal Cannula					
6. Attaches the nasal cannula to the humidifier or the adapter.					
7. Places the nasal prongs in patient's nares, then places the tubing around each ear.					
8. Uses the slide adjustment device to tighten the cannula under patient's chin.					
9. Makes sure that the oxygen equipment is set up correctly and functioning properly before leaving patient's bedside.					
10. Assesses respiratory status before leaving bedside.					

Variation: Face Mask					
11.Gently places the face mask on patient's face, applying it from the bridge of the nose to under the chin.					
12.Secures the elastic band around the back of patient's head, making sure the mask fits snugly but comfortably.					
13.Makes sure that the oxygen equipment is set up correctly and functioning properly before leaving patient's bedside.					
14.Assesses respiratory status before leaving bedside					
15.Gently places the face tent in front of patient's face, making sure that it fits under the chin.					
16.Secures the elastic band around the back of patient's head.					
17.Makes sure that the oxygen equipment is set up correctly and functioning properly before leaving patient's bedside.					
18.Assesses respiratory status before leaving the bedside.					

Body mechanics

Objectives:

Students will be able to:

- Define Body mechanics.
- List the purpose of Body mechanics.
- Explain elements of Body Mechanics.
- Discuss principles of Body Mechanics
- Differentiate between types of assistive devices.
- Demonstrate moving a patient up in bed procedure.
- Demonstrate assisting a patient to a sitting position procedure.
- Demonstrate moving a patient from bed to stretcher procedure.
- Demonstrate Bed to wheelchair transfer procedure.

➢ Definition of Body mechanics:

Body mechanics can be described as the efficient use of one's body to produce motion that is safe, energy conserving and anatomically and physiologically efficient and that leads to the maintenance of a person's body balance and control

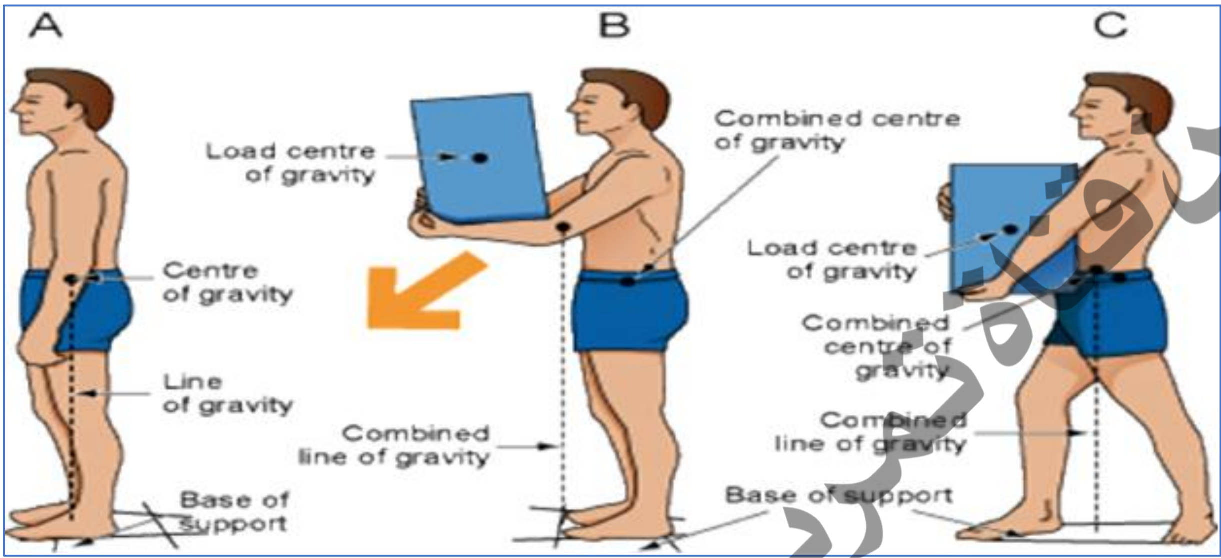
➢ Purpose of body Mechanics:

- Conserve energy.
- Reduce stress and strain on body structures.
- Reduce the possibility of personal injury.
- Produce movements that are safe.

➢ Elements of Body Mechanics:

Body movement requires coordinated muscle activity and neurological integration. It involves the basic elements of **body alignment (posture), balance,**

and coordinated movement. Body alignment and posture bring body parts into position to promote optimal balance and body function.



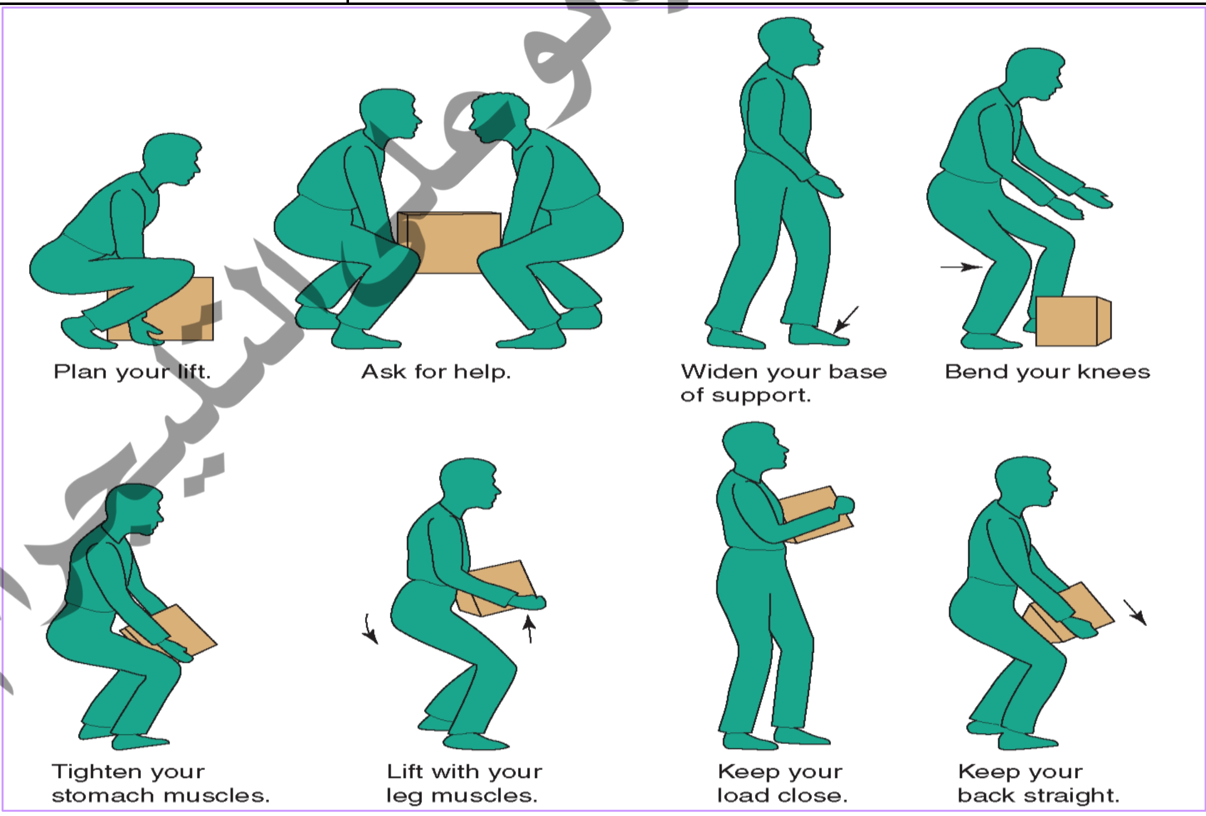
The diagram in Figure demonstrates (A) a well-aligned person whose balance is maintained and whose line of gravity falls within the base of support. Diagram (B) demonstrates how balance is not maintained when the line of gravity falls outside the base of support, and diagram (C) shows how balance is regained when the line of gravity falls within the base of support.

➢ Principles of Body Mechanics:

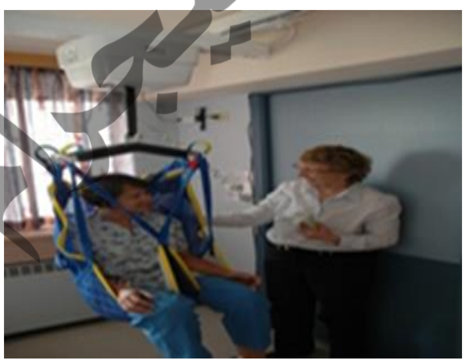
Steps	Principles
Assess the environment.	Assess the weight of the load before lifting and determine if assistance is required.
Plan the move.	Plan the move; gather all supplies and clear the area of obstacles.
Avoid stretching and twisting.	Avoid stretching, reaching, and twisting, which may place the line of gravity outside the base of support.

Ensure proper body stance.	Keep stance (feet) shoulder-width apart. Tighten abdominal, gluteal, and leg muscles in anticipation of the move. Stand up straight to protect the back and provide balance.
Stand close to the object being moved.	Place the weight of the object being moved close to your centre of gravity for balance. Equilibrium is maintained as long as the line of gravity passes through its base of support.  **Hold objects close to your centre of gravity**
Face direction of the movement.	Facing the direction prevents abnormal twisting of the spine.
Avoid lifting.	Turning, rolling, pivoting, and leverage requires less work than lifting. Do not lift if possible; use mechanical lifts as required. Encourage the patient to help as much as possible.
Work at waist level.	Keep all work at waist level to avoid stooping. Raise the height of the bed or object if possible. Do not bend at the waist.

Reduce friction between surfaces.	Reduce friction between surfaces so that less force is required to move the patient.
Bend the knees.	Bending the knees maintains your centre of gravity and lets the strong muscles of your legs do the lifting.
Push the object rather than pull it, and maintain continuous movement.	It is easier to push an object than to pull it. Less energy is required to keep an object moving than it is to stop and start it.
Use assistive devices.	Use assistive devices (gait belt, slider boards, mechanical lifts) as required to position patients and transfer them from one surface to another.
Work with others.	The person with the heaviest load should coordinate all the effort of the others involved in the handling technique.

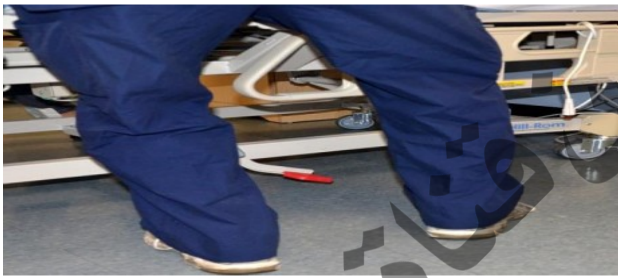
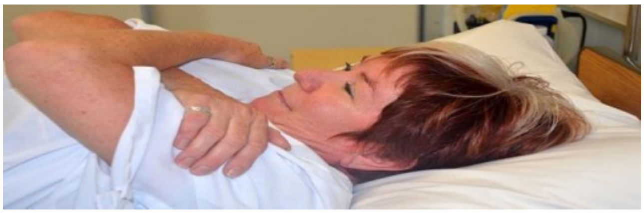


Assistive Devices

Type	Definition
Gait belt or transfer belt 	Used to ensure a good grip on unstable patients. The device provides more stability when transferring patients. It is a 2-inch-wide (5 mm) belt is placed around a patient’s waist.
Slider board or transfer board  **Slider board (red) on a stretcher**	 **Placing a slider board (transfer board) under a patient** A slider board is used to transfer immobile patients from one surface to another while the patient is lying supine. The board allows health care providers to safely move immobile or complex patients.
Mechanical lift 	A mechanical lift is a hydraulic lift, usually attached to a ceiling, used by those who have a medical condition that does not allow them to stand or assist with moving.

Moving a patient up in bed

Steps	Rationale
1. Perform hand hygiene.	- Reduces transmission of microorganisms.
2. Confirm patient ID.	
3. Introduce yourself to patient.	
4.Ensure tubes and attachments are properly placed prior to the procedure	- To prevent accidental removal.
5. Close room curtains or door.	- Provides for patient privacy.
6. Make sure an additional health care provider is available to help with the move.	-This procedure requires two health care providers.
7. Explain to the patient what will happen and how the patient can help.	-Doing this provides the patient with an opportunity to ask questions and help with the positioning.
8. Raise bed to safe working height and ensure that brakes are applied. Health care providers stand on each side of the bed.	-Principles of proper body mechanics help prevent Movement System Impairment (MSI). -Safe working height is at waist level for the shortest health care provider.  **Bed at waist level**
9. Lay patient supine; place pillow at the head of the bed and against the headboard.	-This step protects the head from accidentally hitting the headboard during repositioning.

10. Stand between shoulders and hips of patient, feet shoulder width apart. Weight will be shifted from back foot to front foot.	-This keeps the heaviest part of the patient closest to the center of gravity of the health care providers.  **Feet shoulder width apart**
11. Fan-fold the draw sheet toward the patient with palms facing up.	-This provides a strong grip to move the patient up using the draw sheet.  **Fold sheet with fingers facing upward**
12. Ask patient to tilt head toward chest, fold arms across chest, and bend knees to assist with the movement. Let the patient know when the move will happen.	-This step prevents injury from patient and prepares patient for the move.  **Chin tucked in and arms across chest**
13. Tighten your gluteal and abdominal muscles, bend your knees, and keep back straight and neutral.	-The principles of proper body mechanics help prevent injury.
14. On the count of three by the lead person, gently slide (not lift) the patient	-The principles of proper body mechanics help prevent injury.

up the bed, shifting your weight from the back foot to the front, keeping back straight with knees slightly bent.	 **Facing direction of movement**
15. Replace pillow under head, position patient in bed, and cover with sheets.	-This step promotes comfort and prevents harm to patient.
16. Lower bed, raise side rails as required, and ensure call bell is within reach.	Placing bed and side rails in safe positions reduces the likelihood of injury to patient. Proper placement of call bell facilitates patient's ability to ask for assistance.  **Bed in lowest position, side rail up, call bell within reach**
17. Perform hand hygiene.	-Hand hygiene reduces the spread of microorganisms.
18. Document on the chart with your signature and report any abnormality finding -Documentation provides coordination of care	-Giving signature maintains professional Accountability.

Checklist moving a patient up in bed

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Perform hand hygiene.					
2. Confirm patient ID.					
3. Introduce yourself to patient.					
4.Ensure tubes and attachments are properly placed prior to the procedure					
5. Close room curtains or door.					
6. Make sure an additional health care provider is available.					
7. Explain to the patient what will happen and how the patient can help.					
8. Raise bed to safe working height. Health care providers stand on each side of the bed.					
9. Lay patient supine; place pillow at the head of the bed and against the headboard.					
10. Stand between shoulders and hips of patient, feet shoulder width apart.					
11. Fan-fold the draw sheet toward the patient with palms facing up.					

12. Ask patient to tilt head toward chest, fold arms across chest, and bend knees.					
13. Tighten your gluteal and abdominal muscles, bend your knees, and keep back straight.					
14. On the count of three by the lead person, gently slide the patient up the bed, shifting your weight from the back foot to the front.					
15. Replace pillow under head, position patient in bed, and cover with sheets.					
16. Lower bed, raise side rails as required, and ensure call bell is within reach.					
17. Perform hand hygiene.					
18. Document, and report any abnormality. -Documentation provides coordination of care					

Assisting a patient to a sitting position

Steps	Rationale
1. Perform hand hygiene.	- Reduces transmission of microorganisms.
2. Confirm patient ID.	
3. Introduce yourself to patient.	
4.Ensure tubes and attachments are properly placed prior to the procedure	- To prevent accidental removal.
5. Close room curtains or door.	- Provides for patient privacy.
6. Check physician's order to ambulate and supplies for ambulation if required, and perform an assessment of patient's strength and abilities. -Check physician orders for any restrictions related to ambulation due to medical treatment or surgical procedure.	-Supplies (proper footwear, gait belt, or assistive devices) must be gathered prior to ambulation. Do not leave patient sitting on the side of the bed unsupervised as this poses a safety risk.
7. Explain what will happen and let the patient know how they can help.	-This step provides the patient with an opportunity to ask questions and help with the positioning.
8. Lower bed and ensure brakes are applied.	-This prepares the work environment.
9. Stand facing the head of the bed at a 45-degree angle with your feet apart, with one foot in front of the other. Stand next to the waist of the patient.	-Proper positioning helps prevent back injuries and provides support and balance.

10. Have patient turn onto side, facing toward the caregiver. Assist patient to move close to the edge of the bed.	-This step prepares the patient to be moved.  **Positioning patient on the side of the bed**
11. Place one hand behind patient's shoulders, supporting the neck and vertebrae.	-This provides support for the patient.
12. On the count of three, instruct the patient to use their elbows to push up on the bed and then grasp the side rails, as you support the shoulders as the patient sits up. Shift weight from the front foot to the back foot.	-Do not allow the patient to place their arms around your shoulders. This action can lead to serious back injuries.
13. At the same time as you're shifting your weight, gently grasp the patient's outer thighs with your other hand and help the patient slide their feet off the bed to dangle or touch the floor.	-This step helps the patient sit up and move legs off the bed at the same time.  **Assisting patient into a sitting position**

14. Bend your knees and keep back straight and neutral.	-Use of proper body mechanics helps prevent injury when handling patients.
15. On the count of three, gently raise the patient to sitting position. Ask patient to push against bed with the arm closest to the bed, at the same time as you shift your weight from the front foot to the back foot.	-This allows the patient to help with the process and prevents injury to the health care provider.  **Assist into a sitting position**
16. Assess patient for orthostatic hypotension or vertigo.	-If patient is not dizzy or lightheaded, the patient is safe to ambulate. -If patient becomes dizzy or faint, lay patient back down on bed.
17. Continue with mobilization procedures as required.	-Mobilization helps prevent complications and improves physical function in hospitalized patients.
18. Perform hand hygiene.	-Hand hygiene reduces the spread of microorganisms.
19. Document on the chart with your signature and report any abnormality findings. -Documentation provides coordination of care	-Giving signature maintains professional Accountability.

Checklist assisting a patient to a sitting position

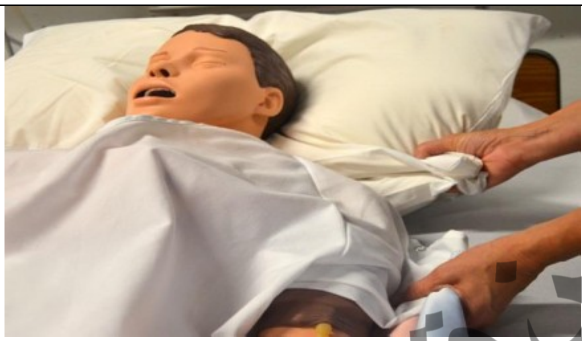
Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Perform hand hygiene.					
2. Confirm patient ID.					
3. Introduce yourself to patient.					
4.Ensure tubes and attachments are properly placed prior to the procedure					
5. Close room curtains or door.					
6. Check physician's order to ambulate, and perform an assessment of patient's strength and abilities.					
7. Explain what will happen and let the patient know how they can help.					
8. Lower bed and ensure brakes are applied.					
9. Stand facing the head of the bed at a 45-degree angle with your feet apart, with one foot in front of the other. Stand next to the waist of the patient.					
10. Have patient turn onto side, facing toward the caregiver. Assist patient to move close to the edge of the bed.					
11. Place one hand behind patient's shoulders, supporting the neck and vertebrae.					
12. On the count of three, instruct the patient to use their elbows to push up on					

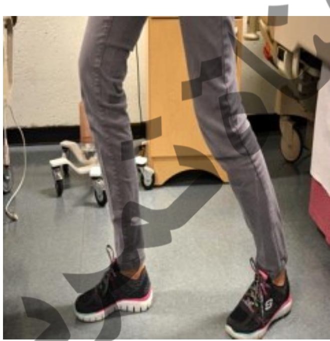
the bed and then grasp the side rails. Shift weight from the front foot to the back foot.					
13. At the same time as you're shifting your weight, gently grasp the patient's outer thighs with your other hand and help the patient slide their feet off the bed to dangle or touch the floor.					
14. Bend your knees and keep back straight.					
15. On the count of three, gently raise the patient to sitting position.					
16. Assess patient for orthostatic hypotension or vertigo.					
17. Continue with mobilization procedures as required.					
18. Perform hand hygiene.					
19. Document and report any abnormality findings. -Documentation provides coordination of care					

Moving a patient from bed to stretcher

Steps	Rationale
1. Perform hand hygiene.	- Reduces transmission of microorganisms.
2. Confirm patient ID.	
3. Introduce yourself to patient.	
4.Ensure tubes and attachments are properly placed prior to the procedure	- To prevent accidental removal.
5. Close room curtains or door.	- Provides for patient privacy.
6. Always predetermine the number of staff required to safely transfer a patient horizontally.	- Three to four health care providers are required for the transfer.
7. Explain what will happen and how the patient can help (tuck chin in, keep hands on chest). -Collect supplies (A slider board and full-size sheet or friction-reducing sheet).	- This step provides the patient with an opportunity to ask questions and help with the transfer.  **Stretcher and slider board**  **Chin tucked in and arms across chest**
8. Raise bed to safe working height. Lower head of bed and side rails.	-Safe working height is at waist level for the shortest health care provider.

-Position the patient closest to the side of the bed where the stretcher will be placed.	-The patient must be positioned correctly prior to the transfer to avoid straining and reaching. -May need additional health care providers to move patient to the side of the bed.
9. Roll patient over and place slider board halfway under the patient, forming a bridge between the bed and the stretcher. -Place sheet on top of the slider board. The sheet is used to slide patient over to the stretcher. -The patient is returned to the supine position. -Patient's feet are positioned on the slider board.	-The slider board must be positioned as a bridge between both surfaces. -The sheet must be between the patient and the slider board to decrease friction between patient and board.  **Placing slider board** -Ensure all tubes and attachments are out of the way.
10. Position stretcher beside the bed on the side closest to the patient, with stretcher slightly lower. Apply brakes. -Two health care providers climb onto the stretcher and grasp the sheet. The lead person is at the head of the bed and will grasp the pillow and sheet.	-The position of the health care providers keeps the heaviest part of the patient near the health care providers' center of gravity for stability.

-The other health care provider is positioned on the far side of the bed, between the chest and hips of the patient, and will grasp the sheet with palms facing up. -The two caregivers on the stretcher grasp the draw sheet using a palm up technique, sitting up tall, and keeping their elbows close to their body and backs straight.	 **Caregiver at the head of the bed**
11. The caregiver on the other side of the bed places his or her hands under the patient's hip and shoulder area with forearms resting on bed.	
12. The designated leader will count 1, 2, 3, and start the move. -The person on the far side of the bed will push patient just to arm's length using a back-to-front weight shift. -At the same time, the two caregivers on the stretcher will move from a sitting-up-tall position to sitting on their heels, shifting their weight from the front leg to the back, bringing the patient with them using the sheet.	-Coordinating the move between health care providers prevents injury while transferring patients. -Using a weight shift from front to back uses the legs to minimize effort when moving a patient.
13. The two caregivers will climb off the stretcher and stand at the side and grasp the sheet, keeping elbows tucked in.	-The step allows the patient to be properly positioned in the bed and prevents back injury to health care providers.

-One of the two caregivers should be in line with the patient's shoulders and the other should be at the hip area. -On the count of three, with back straight and knees bent, the two caregivers use a front-to-back weight shift and slide the patient into the middle of the bed.	 **Caregiver at the head of the bed**  **Weight on front leg**  **Shift weight to back foot**
14. At the same time, the caregiver on the other side slides the slider board out from under the patient.	- This step allows the patient to lie flat on the bed.
15. Replace pillow under head, ensure patient is comfortable, and cover the patient with sheets.	- This promotes comfort and prevents harm to patient.

16. Lower bed and lock brakes, raise side rails as required, and ensure call bell is within reach.	-Placing bed and side rails in a safe position reduces the likelihood of injury to patient. Proper placement of call bell facilitates patient's ability to ask for assistance.
18. Perform hand hygiene.	-Hand hygiene reduces the spread of microorganisms.
19. Document on the chart with your signature and report any abnormality findings. -Documentation provides coordination of care	-Giving signature maintains professional Accountability.

Checklist moving a patient from bed to stretcher

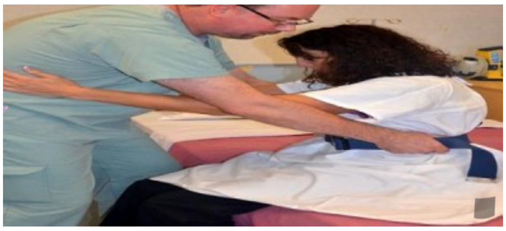
Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Perform hand hygiene.					
2. Confirm patient ID.					
3. Introduce yourself to patient.					
4.Ensure tubes and attachments are properly placed prior to the procedure					
5. Close room curtains or door.					
6. Always predetermine the number of staff required to safely transfer a patient horizontally.					
7. Explain what will happen and how the patient can help (tuck chin in, keep hands on chest).					
8. Raise bed to safe working height. Position the patient closest to the side of the bed where the stretcher will be placed.					
9. Roll patient over and place slider board halfway under the patient, place sheet on top of the slider board. -The patient is returned to the supine position. -Patient's feet are positioned on the slider board.					

10. Position stretcher beside the bed on the side closest to the patient, with stretcher slightly lower. Apply brakes. -Two health care providers climb onto the stretcher and grasp the sheet. The lead person is at the head of the bed and will grasp the pillow and sheet. -The two caregivers on the stretcher grasp the draw sheet using a palm up technique, sitting up tall, and keeping their elbows close to their body and backs straight.					
11. The caregiver on the other side of the bed places his or her hands under the patient's hip and shoulder area with forearms resting on bed.					
12. The designated leader will count 1, 2, 3, and start the move. -The person on the far side of the bed will push patient just to arm's length using a back-to-front weight shift. -At the same time, the two caregivers on the stretcher will move from a sitting-up-tall position to sitting on their heels, shifting their weight from the front leg to the back, bringing the patient with them using the sheet.					

13. The two caregivers will climb off the stretcher and stand at the side and grasp the sheet, keeping elbows tucked in. -On the count of three, with back straight and knees bent, the two caregivers use a front-to-back weight shift and slide the patient into the middle of the bed.					
14. At the same time, the caregiver on the other side slides the slider board out from under the patient.					
15. Replace pillow under head, ensure patient is comfortable.					
16. Lower bed and lock brakes.					
18. Perform hand hygiene.					
19. Document and report any abnormality. -Documentation provides coordination of care					

Bed to wheelchair transfer

Steps	Rationale
1. Perform hand hygiene.	- Reduces transmission of microorganisms.
2. Confirm patient ID.	
3. Introduce yourself to patient.	
4.Ensure tubes and attachments are properly placed prior to the procedure	- To prevent accidental removal.
5. Close room curtains or door.	- Provides for patient privacy.
6. One health care provider is required.	- The patient should be assessed as a 1-person assist.
7. Explain what will happen during the transfer and how the patient can help. - Apply proper footwear prior to ambulation.	- This step provides the patient with an opportunity to ask questions and help with the transfer.
8. Lower the bed and ensure that brakes are applied. -Place the wheelchair next to the bed at a 45-degree angle and apply brakes. If a patient has weakness on one side, place the wheelchair on the strong side.	 **Wheelchair with one leg rest removed**
9. Sit patient on the side of the bed with his or her feet on the floor. Apply the gait belt snugly around the waist (if required). -Place hands on waist to assist into a standing position.	-The patient's feet should be in between the health care provider's feet.  **Patient position prior to standing**

10. As the patient leans forward, grasp the gait belt (if required) on the side the patient, with your arms outside the patient's arms. Position your legs on the outside of the patient's legs. The patient's feet should be flat on the floor.	 **Assist to a standing position using a gait belt**
11. Count to three and, using a rocking motion, help the patient stand by shifting weight from the front foot to the back foot, keeping elbows in and back straight.	 **Weight shift to back leg by health care provider**
12. Once standing, have the patient take a few steps back until they can feel the wheelchair on the back of their legs. Have patient grasp the arm of the wheelchair and lean forward slightly.	- Ensure the patient can feel the wheelchair on the back of the legs prior to sitting down.  **Assist into the wheelchair**
13. As the patient sits down, shift your weight from back to front with bent knees, with trunk straight and elbows slightly bent. Allow patient to sit in wheelchair slowly, using armrests for support.	- This allows the patient to be properly positioned in the chair and prevents back injury to health care providers.  **Transfer to wheelchair**

14. Perform hand hygiene.	-Hand hygiene reduces the spread of microorganisms.
15. Document on the chart with your signature and report any abnormality findings. -Documentation provides coordination of care	-Giving signature maintains professional Accountability.

Checklist Bed to wheelchair transfer

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Perform hand hygiene.					
2. Confirm patient ID.					
3. Introduce yourself to patient.					
4.Ensure tubes and attachments are properly placed prior to the procedure					
5. Close room curtains or door.					
6. One health care provider is required.					
7. Explain what will happen during the transfer and how the patient can help.					
8. Lower the bed and ensure that brakes are applied. Place the wheelchair next to the bed at a 45-degree angle and apply brakes.					
9. Sit patient on the side of the bed with his or her feet on the floor. Place hands on waist.					
10. As the patient leans forward, grasp the gait belt (if required) on the side the patient, with your arms outside the patient's arms.					

11. Count to three and, using a rocking motion, help the patient stand by shifting weight from the front foot to the back foot, keeping elbows in and back straight.					
12. Once standing, patient take a few steps back until they can feel the wheelchair on the back of their legs.					
13. Allow patient to sit in wheelchair slowly, using armrests for support.					
14. Perform hand hygiene.					
15. Document and report any abnormality findings. -Documentation provides coordination of care					

Bed Bath

Objectives:

Students will be able to:

- Define bed bath.
- List purpose of bed bath.
- Differentiate between types of baths.
- Determine equipment's needed during the procedure.
- Demonstrate complete or partial bed bath.

➢ **Definition of bed bath:**

A bath given to client who is in the bed (unable to bath itself)

➢ **Purpose:**

1. To prevent bacteria spreading on skin.
2. To clean the client's body.
3. To stimulate the circulation.
4. To improve general muscular tone and joint.
5. To make client comfort and help to induce sleep.
6. To observe skin condition and objective symptoms.

➢ **Types of Baths:**

- Complete bed bath: Bath administered to totally dependent patient in bed.
- Partial bed bath: Bed bath that consists of bathing only body parts that would cause discomfort if left unbathed such as the hands, face, axilla, and perineal area. Partial bath also includes washing back and providing back rub.
- Sponge bath at the sink: Involves bathing from a bath basin or sink with patient sitting in a chair. Patient is able to perform part of the bath independently. Nurse helps with hard-to-reach areas.

- Tub bath: Involves immersion in a tub of water that allows more thorough washing and rinsing than a bed bath. Patients may require nurse's help.
- Shower: Patient sits or stands under a continuous stream of water. The shower provides more thorough cleaning than a bed bath but can be tiring.
- Disposable bed bath/travel bath: The bag bath contains several soft, nonwoven cotton cloths that are premoistened in a solution of no-rinse surfactant cleaner and emollient. The bag bath offers an alternative because of the ease of use, reduced time bathing, and patient comfort.

- **Equipment:**

1. Basin (2):
    - for without soap (1)
    - for with soap (1)
2. Bucket (2):
    - for clean hot water (1)
    - for waste (1)
3. Jug (1)
4. Soap with soap dish (1)
5. Sponge cloth (2):
    - for wash with soap (1)
    - for rinse (1)
6. Face towel (1)
7. Mackintosh (1)

8. Bath towel (2):
    - (A)for covering over mackintosh (1)
    - (B)for covering over client's body (1)
9. Gauze piece (2-3)
10. Trolley (1)
11. Thermometer (1)
12. Old newspaper
13. Paper bag (2):
    - for clean gauze (1)
    - for waste (1)

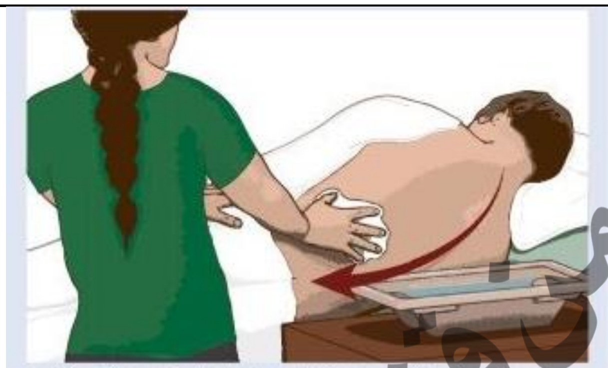
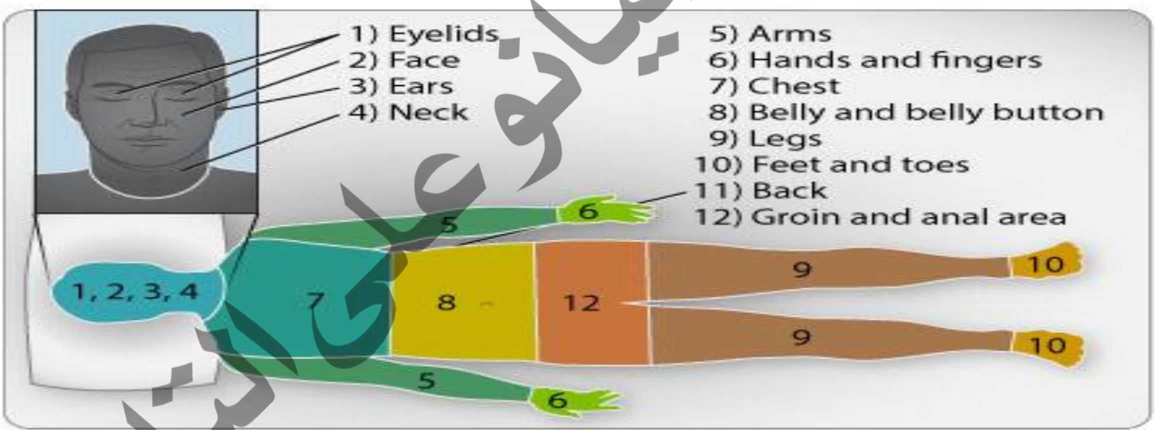
- **Procedure: complete or partial bed bath:**

Steps	Rational
1. Confirm Dr.'s order. Check client identification and condition.	-The bath order may have changed. -In some instances, a bed bath may be harmful for a client, who is in pain, hemorrhaging, or weak.
2. Explain the purpose and procedure to the client. If he or she is alert or oriented, question the client about personal hygiene preferences and ability to assist with the bath.	-Providing information fosters cooperation. -Encourage the client to assist with care and to promote independence.
3. Gather all required equipment to bed-side.	-Organization facilitates accurate skill performance
4. Close the curtain or the door.	-To ensure that the room is warm. -To maintain the client's privacy.
5. Wash your hands- and put-on gloves.	-To prevent the spread of organisms.
6. Prepare warm water.	-Water will cool during the procedure.
7. Remove the client's cloth. Cover the client's body with a top sheet or blanket. If an IV is present on the client's upper extremity, thread the IV tubing and bag through the sleeve of the soiled cloth. Check the IV flow rate.	-Removing the cloth permits easier access when washing the client's upper body. -Be sure that IV delivery is uninterrupted and that you maintain the sterility of the setup.
8. Fill two basins about two-thirds full with warm water.	-Water at proper temperature relaxes him/her and Provides warmth. Water will cool during the procedure.

9. Assist the client to move toward the side of the bed where you will be working.	-Keep the client near you to limit reaching across the bed.
10. Face, neck, ears: 1) Put mackintosh and big towel Ⓐunder the client's body from the head to shoulders. Place face towel under the chin which is also covered the top sheet. 2)Make a mitt with the sponge towel and moisten with plain water. 3) Wash the client's eyes. Cleanse from inner to outer corner. Use a different section of the mitt to wash each eye. 4)Wash the client's face, neck, and ears. Use soap on these areas only if the client prefers. Rinse and dry carefully.	-To prevent the bottom sheet from making wet. -Soap irritates the eyes. -Washing from inner to outer corner prevents sweeping debris into the client's eyes. Using a separate portion of the mitt for each eye prevents the spread of infection. -Soap is particularly drying to the face.  1a. Place a towel under the patient's chin. Wash their eyes, face, neck and ears, check whether they like soap on their face
11.Upper extremities: 1) Move the mackintosh and big towel ○A to under the client's far arm. 2)Uncover the far arm. 3) Fold the sponge cloth and moisten. 4) Wash the far arm with soap and rinse. **Use long strokes:** wrist to elbow→ elbow to shoulder→ axilla→ hand	-To prevent sheet from making wet. - Washing the far side first prevents dripping bath water onto a clean area.

5)Dry by face towel 6) Move the mackintosh and big towel ○A to under the near arm and uncover it 7) Wash, rise, and dry the near arm as same as procedure 4).	- Long strokes improve circulation be facilitating venous return.  1b. Starting with the arm farthest away, wash and dry the top half of the body. Only expose the part of the body being washed 1c. Patients may enjoy soaking their hands and feet in a bowl of water placed on the bed
12.Chest and abdomen: 1) Move the mackintosh and bath towel ○A to under the upper trunk 2) Put another bath towel ○B to over the chest 3) Fold the sponge towel and moisten. 4)Wash breasts with soap and rinse. Dry by the big towel covering. 5) Move the bath towel ○B covering the chest to abdomen. 6)Fold the sponge cloth and moisten. 7)Wash abdomen with soap, rinse and dry 8) Cover the trunk with top sheet and remove the bath towel ○B from the abdomen.	-Mackintosh and bath towel ○A prevent sheet From wetting. -Bath towel ○B provides warmth and privacy.

13. Exchange the warm water.	-Cool bath water is uncomfortable. The water is probably unclean. You may change water earlier, if necessary, to maintain the proper temperature.
14. Lower extremities: 1) Move the mackintosh and bath towel ○A to under the far leg. Put pillow or cushion under the bending knee. Cover the near leg with bath towel ○B. 2) Fold the sponge cloth and moisten. 3)Wash with soap, rinse and dry. **Direction to wash:** from foot joint to knee→ from knee to hip joint 4) Repeat the same procedure as 14.1(- 3) on the near side. 5) Cover the lower extremities with top sheet Remove the cushion, mackintosh and big towel ○A.	-Pillow or cushion can support the lower leg and makes the client comfort. 
15. Turn the client on left lateral position with back towards you.	-To provide clear visualization and easier contact to back and buttocks care.
16.Back and buttocks: 1) Move the mackintosh and big towel ○A under the trunk. 2) Cover the back with big towel ○B. 3) Fold the towel and moisten. Uncover the back.	-Skin breakdown usually occurs over bony prominences. Carefully observe the sacral area and back for any indications.

<table>
<tr><td>4)Wash with soap and rinse. Dry with big towel ○B.
5) Back rub if needed
6) Remove the mackintosh and big towel A.</td><td>
1e. Wash the patient's back and sacrum, moving from top to bottom</td></tr>
<tr><td>17. Return the client to the supine position.</td><td>-To make sustainable position for perineal care.</td></tr>
<tr><td>18. Perineal care.</td><td>-Clean the perineal area to prevent skin irritation and breakdown and to decrease the potential odor.</td></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>19. Assist the client to wear clean cloth.</td><td>-To provide for warmth and comfort.</td></tr>
<tr><td>20.After bed bath:
1) Make the bed tidy and keep the client in comfortable position.</td><td>-These measures provide for comfort and safety</td></tr>
</table>

2) Check the IV flow and maintain it with the speed prescribed if the client is given IV.	-To confirm IV systems going properly and safely.
21. Perform hand wash.	- To prevent the spread of microorganisms.
22. Document on the chart with your signature and report any abnormality findings.	-Documentation provides coordination of care. -Giving signature maintains professional Accountability.

Checklist complete or partial bed bath

Steps	Mark	Trail 1	Trail 2	Trail 3	Comment
1. Confirm Dr.'s order. Check client identification and condition.					
2. Explain the purpose and procedure to the client.					
3. Gather all required equipment to bed- side.					
4. Close the curtain or the door.					
5. Wash your hands- and put-on gloves.					
6. Prepare warm water.					
7. Remove the client's cloth. Cover the client's body with a top sheet or blanket.					
8. Fill two basins about two-thirds full with warm water.					
9. Assist the client to move toward the side of the bed.					
10. Face, neck, ears: 1) Put mackintosh and big towel Ⓐunder the client's body from the head to shoulders. Place face towel under the chin. 2) Make a mitt with the sponge towel and moisten with plain water. 3) Wash the client's eyes. Cleanse from inner to outer corner. 4) Wash the client's face, neck, and ears. Rinse and dry carefully.					

11.Upper extremities: 1) Move the mackintosh and big towel ○A to under the client's far arm. 2) Uncover the far arm. 3) Fold the sponge cloth and moisten. 4) Wash the far arm with soap and rinse. 5)Dry by face towel 6) Move the mackintosh and big towel ○A to under the near arm and uncover it 7) Wash, rise, and dry the near arm.					
12.Chest and abdomen: 1) Move the mackintosh and bath towel ○A to under the upper trunk 2) Put another bath towel ○B to over the chest 3) Fold the sponge towel and moisten. 4) Wash breasts with soap and rinse. Dry by the big towel covering. 5) Move the bath towel ○B covering the chest to abdomen. 6) Wash abdomen with soap, rinse and dry. 7) Cover the trunk with top sheet and remove the bath towel ○B from the abdomen.					
13. Exchange the warm water.					
14. Lower extremities: 1) Move the mackintosh and bath towel ○A to under the far leg. Put pillow or cushion under the					

bending knee. Cover the near leg with bath towel ○B. 2) Fold the sponge cloth and moisten. 3) Wash with soap, rinse and dry. 4) Repeat the same procedure on the near side. 5) Cover the lower extremities with top sheet. Remove the cushion, mackintosh and big towel ○A.					
15. Turn the client on left lateral position with back towards you.					
16.Back and buttocks: 1) Move the mackintosh and big towel ○A under the trunk. 2) Cover the back with big towel ○B. 3) Fold the towel and moisten. Uncover the back. 4) Wash with soap and rinse. Dry with big towel ○B. 5) Back rub if needed. 6) Remove the mackintosh and big towel A.					
17. Return the client to the supine position.					
19.perineal care -Documentation provides coordination of care					
19. Assist the client to wear clean cloth.					
20. Make the bed tidy and keep the client in comfortable position.					
21. Perform hand wash.					
22. Document and report any abnormality findings.					

Bed making

Objectives

Student will be able to

- Define bed making
- Identify nursing consideration for bed making
- Identify indication for occupied bed making
- Determine equipment's needed during bed making
- Demonstrate the intervention for bed making

Introduction

Bed making is a procedure the nurses must be oriented to evaluation making an occupied and unoccupied bed separately from giving a bed bath. If linen change is done at the same time as the bed bath

Making an occupied bed

Definition

When the patient cannot get out of bed, the linens may need to be changed with the patient still in the bed.

Nursing consideration:

1- Assess the patient's preferences regarding linen changes.
2- Assess for any precautions or activity restrictions for the patient.
3- The bed linens are applied without injury to the patient or nurse.
4- Patient verbalizes feelings of increased comfort.

Indication

Patient unable to move outside the bed (as in paralysis, coma, patient on ventilator, fracture).

Equipment

1. One large flat sheet
2. One fitted sheet

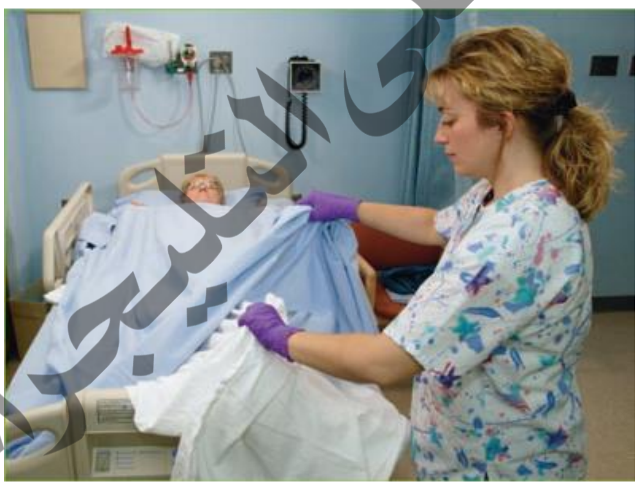
3. Draw sheet (optional)
4. Blankets
5. Bedspread
6. **Pillowcases**
7. Linen hamper or bag
8. Bedside chair
9. Protective pad (optional)
10. Disposable gloves
11. Additional PPE, as indicated

Procedure

Steps	**Rational**
1. Check health care record for limitations on patient's physical activity.	This facilitates patient cooperation, determines level of activity, and promotes patient safety.
2. Perform hand hygiene. 3. Put on PPE, as indicated.	Hand hygiene and PPE prevent the spread of microorganisms. PPE is required based on transmission precautions.
4. Identify the patient. Explain what you are going to do.	Patient identification validates the correct patient and correct procedure. Discussion and explanation allay anxiety and pre- pare the patient for what to expect.
5. Place equipment on over-bed table within reach. a. Moves furniture to allow access to the bed. b. Positions the linen hamper for easy access.	Organization facilitates performance of task.
6. Close curtains around bed and close the door to the room, if possible	This ensures the patient's privacy.
7. Adjust the bed to a comfortable working height.	Having the bed at the proper height prevents back and muscle strain.
8. Lowers the side rail nearest the nurse. Place bed in flat position unless contraindicated.	Having the mattress flat makes it easier to prepare a wrinkle- free bed.

9. Checks that no tubes are entangled in the bed linens.	Disconnecting tubes from linens prevents discomfort and accidental dislodging of the tubes.
10. Place a bath blanket over the patient. Have patient hold on to bath blanket while you reach under it and remove top linens 	The blanket provides warmth and privacy. Placing linens directly into the hamper helps prevent the spread of microorganisms.
11. If possible, and another person is available to assist, grasp mattress securely and shift it up to head of bed.	This allows more foot room for the patient.
12. Assist patient to turn toward opposite side of the bed, and reposition pillow under patient's head.	This allows the bed to be made on the vacant side.
13. Loosen all bottom linens from head, foot, and side of bed.	This facilitates removal of linens.
14. Roll soiled linens as close to the patient as possible. 	This makes it easier to remove linens when the patient turns to the other side.
15. Use clean linen and make the near side of the bed. Place the bottom sheet with its center fold in the center of the bed.	Opening linens on the bed reduces strain on the nurse's arms and diminishes the spread of

 Open the sheet and roll the center, positioning it under the old linens. 	microorganisms. Centering the sheet ensures sufficient coverage for both sides of the mattress. Positioning under the old linens makes it easier to remove linens.
16. Raise side rail. Assist patient to roll over the folded linen in the middle of the bed toward you. Reposition pillow and bath blanket or top sheet. Move to other side of the bed and lower side rail.	This ensures patient safety. The movement allows the bed to be made on the other side. The bath blanket provides warmth and privacy
17. Loosen and remove all bottom linen. Discard soiled linen in laundry bag or hamper. Do not place on floor or furniture. Do not hold soiled linens against your uniform.	Placing linens directly into the hamper helps prevent the spread of microorganisms. The floor is heavily contaminated; soiled linen will further contaminate furniture. Soiled linen contaminates the nurse's

	uniform, and this may spread organisms to another patient
18. Ease clean linen from under the patient. Pull the bottom sheet and secure at the corners at the head and foot of the mattress.	This removes wrinkles and creases in the linens, which are uncomfortable to lie on.
19. Assist patient to turn back to the center of bed. Remove pillow and change pillowcase.	Opening linens by shaking them causes organisms to be carried on air currents.
20. Apply top linen, sheet, and blanket, if desired, so that it is centered. Fold the top linens over at the patient's shoulders to make a cuff 	This allows bottom hems to be tucked securely under the mattress and provides for privacy

21. Return patient to a position of comfort. Remove your gloves. Raise side rail and lower bed. Reattach call bell. Positions the bedside table and overbed table within patient's reach. 	Promotes patient comfort and safety. Removing gloves properly reduces the risk for infection transmission and contamination of other items.
22. Dispose of soiled linens according to agency policy.	Deters the spread of microorganisms.
23. Remove any other PPE, if used. Perform hand hygiene.	Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene prevents the spread of microorganisms.

Procedure Checklist

STEPS	Mark	Trial 1	Trial 2	Trial 3	Comment
1. Check chart for limitations on patient's physical activity					
2. Assemble equipment and arrange on bedside chair in the order the items will be used.					
3. Perform hand hygiene. Put on PPE, as indicated.					
4. Identify the patient. Explain what you are going to do.					
5. Keep the patient's privacy					
6. Adjust the bed to a comfortable working height.					
7. Lowers the side rail nearest the nurse. Place bed in flat position unless contraindicated.					
8. Checks that no tubes are entangled in the bed linens.					
9. Place a bath blanket over the patient. Have patient hold on to bath blanket while you reach under it and remove top linens					
10. If possible, and another person is available to assist, grasp mattress securely and shift it up to head of bed.					
11. Assist patient to turn toward opposite side of the bed, and reposition pillow under patient's head.					

12.Loosen all bottom linens from head, foot, and side of bed.					
13.Fan-fold soiled linens as close to patient as possible.					
14.Use clean linen and make the near side of the bed. Place the bottom sheet with its center fold in the center of the bed. Open the sheet and fan-fold to the center, positioning it under the old linens. Pull the bottom sheet over the corners at the head and foot of the mattress.					
15.Raise side rail. Assist patient to roll over the folded linen in the middle of the bed toward you. Reposition pillow and bath blanket or top sheet. Move to other side of the bed and lower side rail.					
16.Loosen and remove all bottom linen. Discard soiled linen in laundry bag or hamper. Do not place on floor or furniture. Do not hold soiled linens against your uniform.					
17.Ease clean linen from under the patient. Pull the bottom sheet and secure at the corners at the head and foot of the mattress.					
18.Assist patient to turn back to the center of bed. Remove pillow and change pillowcase.					

19.Apply top linen, sheet, and blanket, if desired, so that it is centered. Fold the top linens over at the patient's shoulders to make a cuff					
20.Secure top linens under foot of mattress and miter corners. Loosen top linens over patient's feet by grasping them in the area of the feet and pulling gently toward foot of bed.					
21.Return patient to a position of comfort. Remove your gloves. Raise side rail and lower bed. Reattach call bell.					
22.Dispose of soiled linens according to agency policy.					
23.Remove any other PPE, if used. Perform hand hygiene					

Making bed (unoccupied)

Steps	Rational
1. Perform hand hygiene. Put on PPE, as indicated.	Hand hygiene and PPE prevent the spread of microorganisms. PPE is required based on transmission precautions.
2. Explain to the patient what you are going to do and the reason for doing it. Assists patient to the chair; provides a robe/blanket if needed.	Discussion and explanation allay anxiety and pre- pare the patient for what to expect.
3. Prepares the environment: a. Moves furniture as needed. b. Places the linen hamper for convenient access.	Organization facilitates performance of task.
4. Adjust the bed to a comfortable working height, and lowers side rails.	Having the bed at the proper height prevents back and muscle strain.
5. Disconnect call bell or any tubes from bed linens.	Disconnecting devices prevents damage to the devices.
6. Fold reusable linens, such as sheets, blankets, or spread, in place on the bed in fourths and hang them over a clean chair	Folding saves time and energy when reusable linen is replaced on the bed. Folding linens while they are on the bed reduces strain on the nurse's arms. Some agencies change linens only when soiled.

7. roll all the soiled linen inside the bottom sheet and place directly into the laundry hamper 	Rolling soiled linens snugly and placing them directly into the hamper helps prevent the spread of microorganisms.
8. Remove your gloves, unless indicated for transmission precautions. Place the bottom sheet with its center fold in the center of the bed. Open the sheet and fan-fold to the center 	Gloves are not necessary to handle clean linen. Removing gloves properly reduces the risk for infection transmission and contamination of other items. Opening linens on the bed reduces strain on the nurse's arms and diminishes the spread of microorganisms.
9. If using, place the drawsheet with its center fold in the center of the bed and positioned so it will be located under the patient's midsection. 	If the patient soils the bed, drawsheet and pad can be changed without the bottom and top linens on the bed.

10. Pull the bottom sheet over the corners at the head and foot of the mattress.	Making the bed on one side and then completing the bed on the other side saves time. Having bottom linens free of wrinkles reduces patient discomfort.
11. Move to the other side of the bed to secure bottom linens. Pull the bottom sheet tightly	This removes wrinkles from the bottom linens, which can cause patient discomfort and promote skin breakdown.
12. Place the top sheet on the bed with its center fold in the center of the bed	Opening linens by shaking them spreads organisms into the air.
13. Tuck the top sheet and blanket under the foot of the bed on the near side. 	This saves time and energy and keeps the top linen in place.
14. Move to the other side of the bed and follow the same procedure for securing top sheets under the foot of the bed and making a cuff 	Working on one side of the bed at a time saves energy and is more efficient.

15. Place the pillows on the bed. Open each pillowcase in the same manner as you opened other linens.	Covering the pillow while it rests on the bed reduces strain on the nurse’s arms and back.
	
16. Secure the signal device on the bed, according to agency Policy 	The patient will be able to call for assistance as necessary. Promotes patient comfort and safety.
17. Assists patient back to bed. Raise side rail and lower bed.	Promotes patient comfort and safety
18. Dispose of soiled linen according to facility policy.	Deters the spread of microorganisms.
19. Remove any other PPE, if used. Perform hand hygiene.	Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene prevents the spread of microorganisms.

Procedure Checklist

PROCEDURE STEPS	Mark	Trial 1	Trial 2	Trial 3	comment
1. Perform hand hygiene. Put on PPE, as indicated.					
2. Explain to the patient what you are going to do and the reason for doing it. Assists patient to the chair; provides a robe/blanket if needed.					
3. Prepares the environment: a. Moves furniture as needed. b. Places the linen hamper for convenient access.					
4. Adjust the bed to a comfortable working height, and lowers side rails.					
5. Disconnect call bell or any tubes from bed linens.					
6. Fold reusable linens, such as sheets, blankets, or spread.					
7. roll all the soiled linen inside the bottom sheet and place directly into the laundry hamper					
8. Remove your gloves, Place the bottom sheet with its center fold in the center of the bed.					
9. If using, place the drawsheet with its center fold in the center of the bed.					
10. Pull the bottom sheet over the corners at the head and foot of the mattress.					

11. Move to the other side of the bed to secure bottom linens.					
12. Place the top sheet on the bed with its center fold in the center of the bed					
13. Tuck the top sheet and blanket under the foot of the bed on the near side.					
14. Move to the other side of the bed and follow the same procedure.					
15. Place the pillows on the bed. Open each pillowcase in the 16. same manner as you opened other linens.					
17. Secure the signal device on the bed, according to agency 18. Policy					
19. Assists patient back to bed. Raise side rail and lower bed.					
20. Dispose of soiled linen according to facility policy.					
21. Remove any other PPE, if used. Perform hand hygiene.					

Medication administration

Objectives

Student will be able to

- Define Medication administration.
- Discuss National Patient Safety Goals for medication administration.
- List the ten rights of medication administration.
- Differentiate among different types of routes of medication administration.
- Discuss Medication Preparation measures.
- Demonstrate Medication administration procedure for each route

➢ **Medication administration** means the direct application of a prescribed medication whether by injection, inhalation, ingestion, or other means, to the body of the resident by an individual legally authorized to do so.

➢ A **medication** is a substance administered for the diagnosis, cure, treatment, or relief of a symptom or for prevention of disease.

The Joint Commission 2023 Hospital National Patient Safety Goals: Implications or Medication Administration

1. Identify patient correctly. Use at least two patient identifiers (neither can be patient's room number) when providing care, treatment (e.g., medications), or services.
2. Improve the effectiveness of communication among caregivers.
3. Verbal or telephone orders require a verification "read-back" of the complete order or test result by the person receiving the order/test result.
4. Standardize a list of abbreviations, acronyms, symbols, and dose designations that are not to be used throughout an organization.
5. Improve the safety of using medications.

6. Identify and at a minimum annually review a list of look-alike/ sound-alike drugs used by the organization.
7. Before a procedure, label all medications and medication containers (e.g., syringes) that are not labeled. Do this in areas where medicines and supplies are set up. Labels include drug name, strength, amount, expiration date when not used within 24 hours, and expiration time when expiration occurs in less than 24 hours.
8. Take extra care with patients who take anticoagulants. Use only oral unit-dose products and premixed infusions. When heparin is administered intravenously and continuously, use programmable infusion pumps.
9. Maintain and communicate accurate patient medication information.
10. Accurately and completely reconcile medications across the continuum of care.
11. There is a process for comparing the patient's current medications with those ordered for the patient while under the care of the health care organization.
12. Communicate a complete list of the patient's medications to the next provider of service when a patient is referred or transferred to another setting, service, or level of care.
13. encourage patients' active involvement in their own care as a patient safety strategy

Medication Preparation

1. **Interpreting Medication Labels** Medication labels include the trade name of the drug in large letters, the generic name in smaller letters, the form of the drug, the dosage, the expiration date, the lot number, and the name of the manufacturer.
2. **Minimize distractions** during medication preparation (e.g., discussion with staff, phone call, pager), and do not perform other tasks while preparing a medication. Do not allow interruptions.

3. **Read the label on the medication container** and compare it with the medication administration record (MAR) **at least 3 times**: before removing the container from the supply drawer, when placing the medication in an administration syringe, and just before administering the medication to the patient.
4. **Double-check all calculations** and other high-risk medication administration processes (e.g., insulin, patient-controlled analgesia) and verify with another nurse.
5. **Use good hand-hygiene technique.** Avoid touching tablets and capsules. Use sterile technique for parenteral medications. Wear clean gloves when administering parenteral medications and certain topical medications.
6. **Administer only those medications that you personally prepare.** Do not ask another person to administer medications that you prepare. Keep medications secure.
7. **When preparing medications, be sure that the label is clear and legible and that the drug is mixed properly**; has not changed in color, clarity, or consistency; and has not expired.
8. **Keep tablets and capsules in their wrappers** and open them at the patient's bedside. This allows you to review each medication with the patient. If a patient refuses medication, there will be no question which one was withheld.

Routes of Drug Administration

- Oral	- Topical
- Parenteral	- Intradermal
- Subcutaneous	- Intramuscularly
- Intravenous	- Rectal
- Vaginal	- Inhalation

Note

1. Remember the 10 R's

1. Right patient
2. Right medication
3. Right route
4. Right dose
5. Right time
6. Right documentation
7. Right History and Assessment.
8. Right to Refuse.
9. Right Drug-Drug Interaction and Evaluation.
10. Right Education and Information.

2. Always keep the bottle tightly closed.

3. Clean and keep the label of the bottle clear.

4. Keep medication away from light.

5. Cheek their expiration date.

6. Keep the rim of the bottle clean.

7.Give your undivided attention to your work while preparing and giving medications.

8. Make sure that a graduate nurse checks some potent drugs.

9. Never give medications from unlabeled container

10. Never return a dose once poured from the bottle.

11.Check your patient's vital sign may be necessary before and after administrating some drugs e.g., digitals.

12. Never give medicine that someone poured or drawn.

13. Never leave medicine at bed side of a patient and within reach of the children

➢ Oral Administration

Definition: Oral medication is drug administered by mouth

Indication

a. When local effects on GI tract are desired
b. When prolonged systemic action is desired

Contra indications:

1. for a patient with nausea & vomiting, unconscious patients.
2. When digestive juices inactivate the effect of the drug.
3. When there is inadequate absorption of the drug, which leads to inaccurate determination of the drug absorbed.
4. When the drug is irritating to the mucus membrane of the alimentary canal.

Equipment

- Tray
- Towel
- A bowl of water for used mediation cup
- Measuring spoon
- A Jug of water (boiled water)
- Chart and medication card
- Ordered medication
- Straw if necessary

Procedure

Steps	Rational
1. Hand washing	To prevent cross infection
2. Explain procedures to the patients	To decrease patient fear &gain cooperation
3. Prepare your tray and take it to the patient's room	To save time and effort
4. Begin by checking the order& read the label 3 times	To ensure that is the right patient
5. Place solution and tablets in a separate container.	To prevent contamination
6. If suspension, shake the bottle well before pouring	To facilitate drug distribution
7. Take it to the patient bedside	To prevent mistakes
8. Keep the medication in site at all times	To maintain patient rights
9. Identify the patient carefully using all identification variables. (Pt's name, bed number…)	To be sure that patient takes medication.
10.Remain with the patient until each medicine is swallowed	
11. Offer additional fluid as necessary unless contra-indicated	To facilitate absorption
12.Record the medication given, refused or omitted immediately.	To maintain safety to both the nurse & the patient
13.Take care of the equipment & return them to their proper places. 14.Wash your hands.	To prevent cross infection

Procedure Checklist

STEPS	Mark	Trial 1	Trial 2	Trial 3	comments
1. Hand washing					
2. Explain procedures to the patients					
3. Prepare your tray and take it to the patient's room					
4. Begin by checking the order& read the label 3 times					
5. Place solution and tablets in a separate container.					
6. If suspension, shake the bottle well before pouring					
7. Take it to the patient bedside					
8. Keep the medication in site at all times					
9. Identify the patient carefully using all identification variables. (Pt's name, bed number…)					
10. Remain with the patient until each medicine is swallowed					
11. Offer additional fluid as necessary unless contra-indicated					
12. Record the medication given, refused or omitted immediately.					
13. Take care of the equipment & return them to their proper places.					
14. Wash your hands.					

➢ II. Parenteral Drug Administration

1- Intradermal Injection

Definition: It is an injection given into the dermal layer of the skin

Purpose

For diagnostic purpose

a. Fine test (Mantoux test)

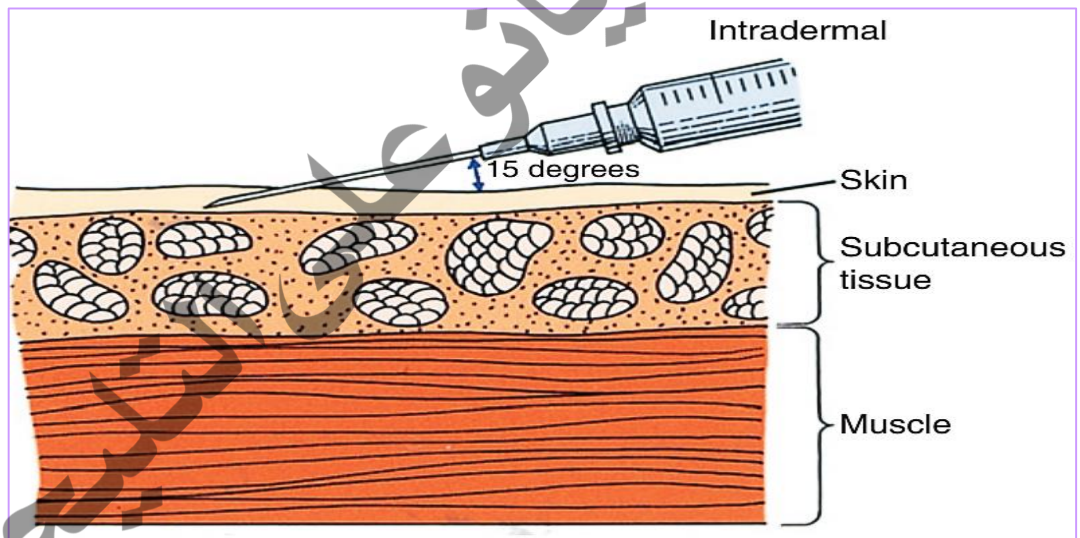
b. Allergic reaction

For therapeutic purpose

c. Intradermal injection may also be given like in vaccination

Site of Injection

- The inner part of the forearm (midway between the wrist and elbow.
- Upper arm, at deltoid area for BCG vaccination



Intradermal needle tip inserted into dermis.

Equipment

- Tray
- Syringe & needle (sterile)
- Receiver

- Drug (to be injected)
- File
- Alcohol swab
- Marking pen
- Water in the bowel to rinse syringe and needle

Procedure

Steps	Rational
1. Wash hands and put on clean gloves.	Reduces the transmission of microorganisms.
2. in the inpatient setting, close door or curtains around bed and keep gown or sheet draped over body. In the outpatient setting, close door to exam or treatment room. Identify client.	Provides privacy. Assures medication is given
3. Select injection site. • Inspect skin for bruises, inflammation, edema, masses, tenderness, and sites of previous injections. • Forearm site should be 3–4 finger widths below antecubital space and one hand width above wrists on inner aspect of forearm.	Injection site should be free of lesions. Repeated daily injections should be rotated. Assures a clear site for interpreting results.
4. Select 1⁄4- to 5⁄8-inch 25- to 27-gauge needle	Ensures that needle will be injected into the intradermis.
5. Assist client into a comfortable position. Forearm site: Relax the arm with elbow and fore arm extended on a flat surface. Distract client by talking about an interesting subject.	Relaxation minimizes discomfort. Distraction reduces anxiety.
6. Use antiseptic swab in a circular motion to clean skin at site.	Circular motion and mechanical action of swab remove secretions containing microorganisms

7. While holding the swab between fingers of nondominant hand, pull cap from needle.	Swab remains accessible during procedure. Prevents contamination of needle.
8. Administer injection: • With nondominant hand, stretch skin over site with forefinger and thumb. • Insert needle slowly at a 5- to 15-degree angle, bevel up, until resistance is felt; then advance to no more than 1⁄8 inch below the skin. The needle tip should be seen through the skin. •Do not aspirate. Slowly inject the medication. Resistance will be felt. • Note a small bleb, like a mosquito bite, forming under the skin surface. 	•Needle penetrates tight skin easier than loose skin. •Ensures needle tip is in the dermis. •Dermal layer is tight and does not expand easily when fluid is injected. •Indicates the medication was deposited in the dermis
9. Withdraw the needle while applying gentle pressure with the antiseptic swab.	Supporting tissue around injection site minimizes discomfort.
10. Do not massage the site.	Prevents medication from being dispersed into the tissue and altering test results.
11. Assist the client to a comfortable position.	Promotes comfort.
12. Discard the uncapped needle and syringe in a safe receptacle.	Decreases risk of needle stick.
13. Remove gloves and wash hands.	Reduces transmission of organisms.
14. Document about the procedure	To avoid farther complication

Procedure Checklist

STEPS	Mark	Trial 1	Trial 2	Trial 3	Comments
1. Identifies the patient according to agency policy using two identifiers; perform hand hygiene, safety, privacy, body mechanics, and documentation.					
2. Draws up medication from vial.					
3. Selects the site for injection (usual sites are the ventral surface of the forearm and upper back; upper chest may also be used).					
4. Assists patient to a comfortable position.					
5. Put on clean procedure gloves.					
6. Cleanses the injection site with an alcohol prep pad, or another antiseptic swab. circles from the center of the site outward.					
7. Holds the syringe between the thumb and index finger of the dominant hand parallel to skin; removes the needle cap.					
8. Using the nondominant hand, holds the patient's skin taut					
9. holds the syringe in the dominant hand with the bevel up and parallel to patient's skin at a 5° to 15° angle.					
10. Inserts the needle slowly and advances approximately 3 mm (⅛ in.) so that the entire					

bevel is covered. The bevel should be visible just under the skin.					
11. Does not aspirate. Holds syringe stable with nondominant hand.					
12. Releases the taut skin and slowly injects the solution. A pale wheal, about 6–10 mm (¼ in.) in diameter, should appear over needle bevel.					
13. Removes the needle from the skin, engages the safety needle device, and disposes in a biohazard puncture-proof container.					
14. Gently blots any blood with a dry gauze pad. Does not rub or cover with an adhesive bandage.					
15. With a pen, draws a 1-in. circle around the bleb/ wheal.					
16. Remove gloves and wash hands.					
17. Document about the procedure					

2- Sub - Cutaneous Injection

Definition: Injecting of drug under the skin in the sub- cutaneous tissue, (under the dermis)

Purpose:

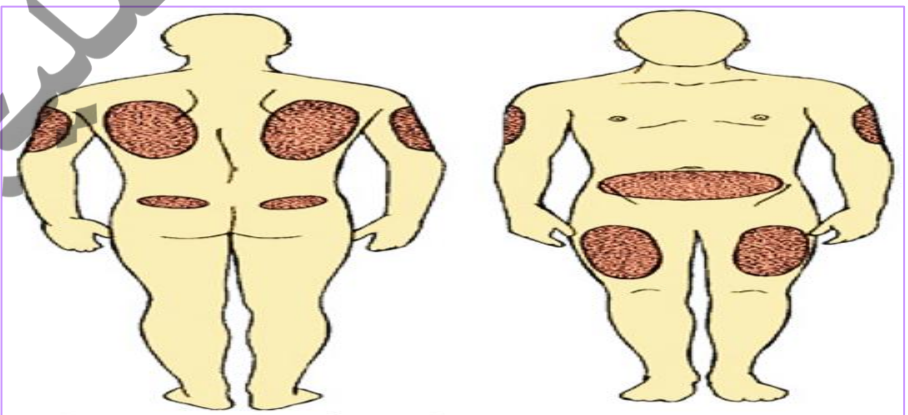
- To obtain quicker absorption than oral administration
- When it is impossible to give medication orally

Equipment

- Tray
- Alcohol swabs
- File
- Receiver
- Disposing box
- Sterile syringe & needle (disposable)
- Medication
- Medication card and patient chart
- Water in a bowel

Site of Injection

- Outer part of the upper arm
- The abdomen below the costal margin to the iliac crest.
- The anterior aspect of the thigh



Common sites for subcutaneous injections.

Procedure

Steps	Rational
1. Wash hands and put on clean gloves.	1. Reduces the number of microorganisms.
2. Close door or curtains around bed and keep gown or sheet draped over body. Identify client.	2.Provides privacy. Assures medication is given to right client.
3. Select injection site. •Inspect skin for bruises, inflammation, edema, masses, tenderness, and sites of previous injections. • Use anatomic landmarks. 	3. Injection site should be free of lesions. • Repeated daily injections should be rotated. • Avoids injury to underlying nerves, bone, or blood vessels.
4. Select needle size: • Measure skinfold by grasping skin between thumb and forefinger. • Be sure needle is one-half the length of the skinfold from top to bottom.	4. Ensures that needle will be injected into subcutaneous tissue
5. Assist client into a comfortable position: • Relax the arm, leg, or abdomen. • Distract client by talking about an interesting subject or explaining what you are doing step by step.	5. Relaxation minimizes discomfort. Distraction reduces anxiety.

6. Use antiseptic swab to clean skin at site. 	6. Circular motion and mechanical action of swab remove microorganisms.
7. While holding swab between fingers of nondominant hand, pull cap from needle.	7. Swab remains accessible during procedure. Prevents contamination of needle.
8. Administer injection: • Hold syringe between thumb and forefinger of dominant hand like a dart. • Pinch skin with nondominant hand.  • Inject needle quickly and firmly (like a dart) at a 45- to 90-degree angle.  • Release the skin.	8. • Quick, smooth injection is easier with proper position of syringe. •Needle penetrates tight skin easier than loose skin. Pinching skin elevates subcutaneous tissue. • Quick, firm injection minimizes discomfort.

• Grasp the lower end of the syringe with nondominant hand and position dominant hand to the end of the plunger. Do not move the syringe. • Pull back on the plunger to ascertain that the needle is not in a vein. If no blood appears, slowly inject the medication. (Aspiration is contraindicated with some medications; check with the pharmacy if you are unclear.)	•Injection requires smooth manipulation of syringe parts. Movement of syringe may cause discomfort. • Aspiration of blood indicates intravenous placement of needle so procedure may have to be abandoned.
9. Quickly withdraw the needle while applying pressure with the antiseptic swab. Do not push down on the needle with the swab while withdrawing it, as this will cause more pain.	9.Supporting tissue around injection site minimizes discomfort.
10. Gently massage the site. Some medications should not be massaged. Ask the pharmacy if you are unclear.	10. Stimulates circulation and improves drug distribution and absorption.
11. Assist the client to a comfortable position.	11. Promotes comfort
12. Discard the uncapped needle and syringe in a disposable needle receptacle.	12. Decreases risk of needle stick.
13. Remove gloves and wash hands.	13.Reduces transmission of microorganisms.
14.Document the date, time, dose, route, site of injection, and signature or initials.	14. To obtain base line data
Note. If repeated injections are given, the nurse should rotate the site of injection so that each succeeding injection is about 5 cm away from the previous one.	

Procedure Checklist

STEPS	Mark	Trial 1	Trial 2	Trial 3	Comments
1. Hand washing					
2. Take equipment to the patient’s bed side					
3. Explain the procedure to the patient					
4. Draw your medication					
5. Expel the air from the syringe					
6. Clean the site (usually it is in upper arms, thighs or abdomen)					
7. Grasp the area between your thumb & forefinger to tense it.					
8. Insert the needle elevate about 45 angles.					
9. Pierce the skin quickly & advance the needle					
10. Aspirate to determine that the needle has not entered a blood vessel					
11. Inject the drug slowly.					
12. After injecting withdraw the needle and apply gentle pressure to site. with alcohol swab.					
13. document the amount and time of administration immediately.					
14. Take care of the equipment- wash, sterilize and return to its place					
15. Watch for undesired reaction (side effect of the drug) etc.					

3- Intra Muscular Injection (IM)

Definition: It is an introduction of a drug into a body's system via the muscles.

Purpose

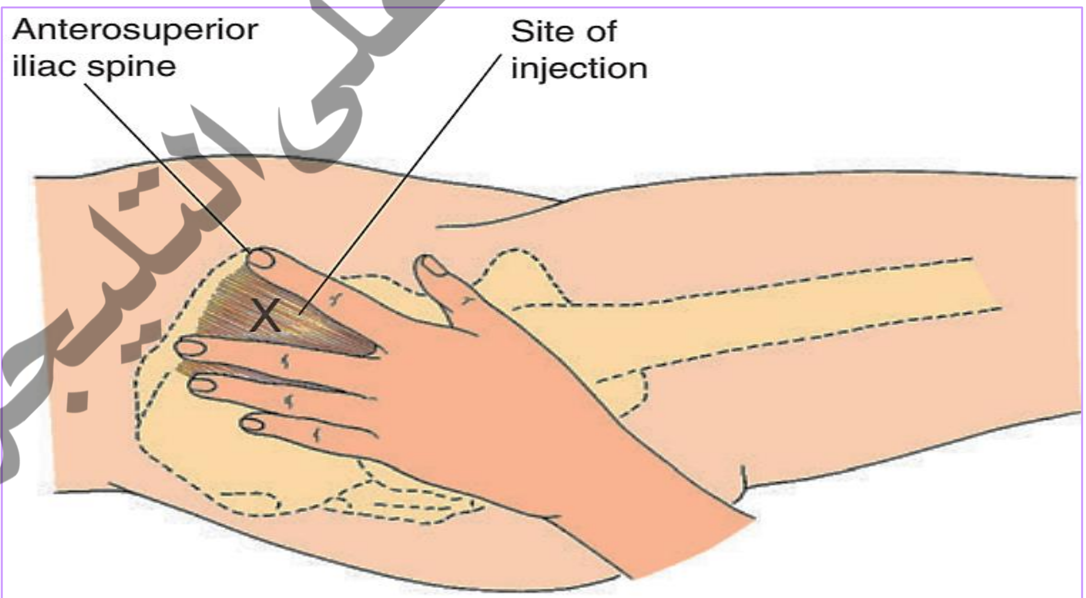
- To obtain quick action next to the intra- venous route
- To avoid an irritation from the drug if given through another route.

Equipment

- Tray
- Ordered drug (ampoule, vial)
- Sterile syringes and needle in a container
- Alcohol swab
- Receiver
- A bowl of water for used syringes and needle
- File
- Sterile jar with sterile forceps
- Chart

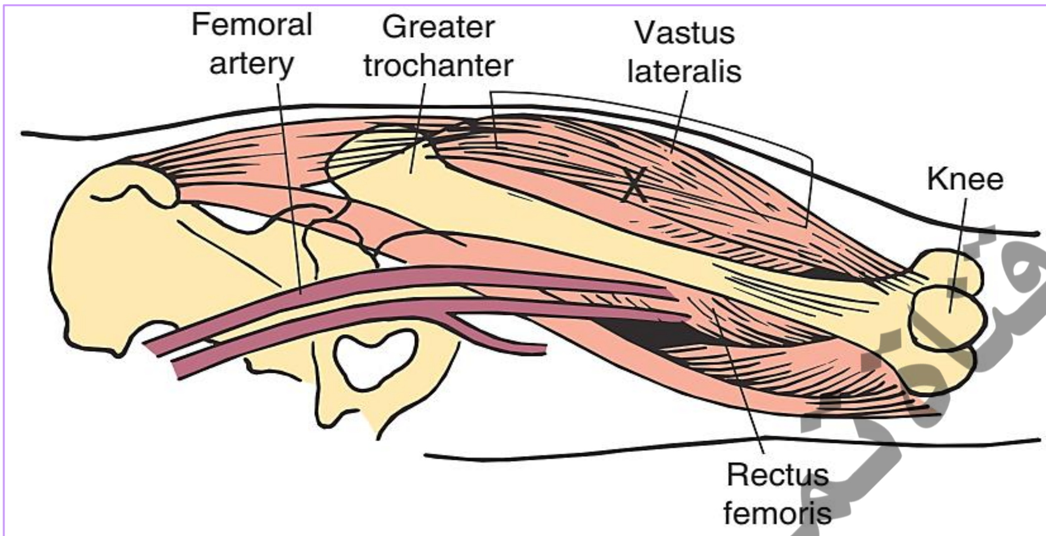
Sites for I.M. Injection

- Ventrogluteal muscle
- Dorsolateral muscle



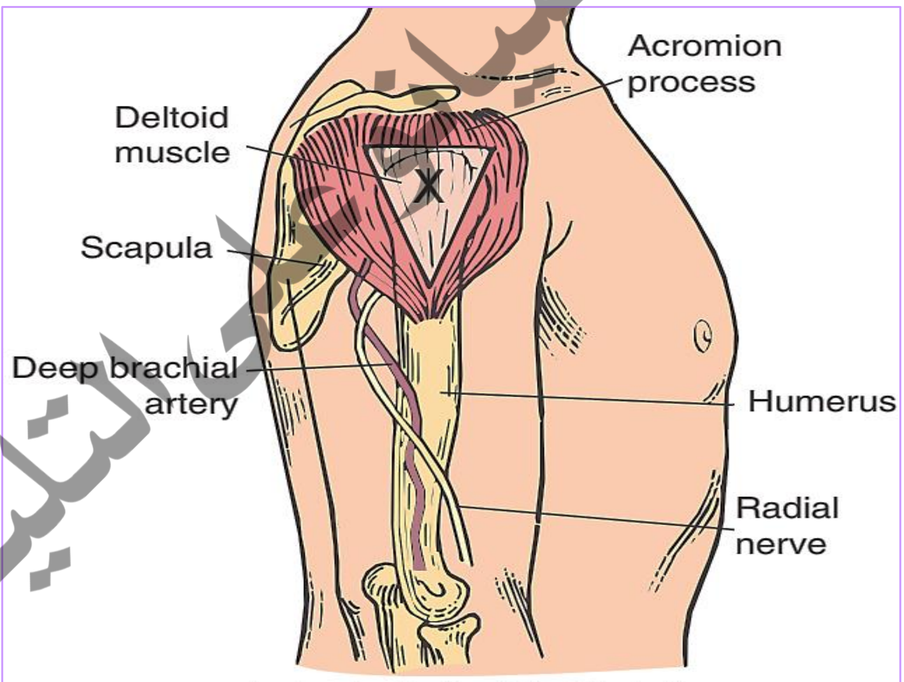
Anatomic site for ventrogluteal injection.

• Vastus Lateralis



Anatomic site for vastus lateralis injection.

• Deltoid muscle



Anatomic site for deltoid injection.

Procedure

Steps	Rational
1. Wash hands and put on clean gloves.	1.Reduces the number of microorganisms.
2. Close door or curtains around bed and keep gown or sheet draped over body. Identify client	2.Provides privacy.
3.Identify patient using two identifiers (e.g., name and birthday or name and account number) according to agency policy.	3.Assures medication is given to the right client.
4. Select injection site. •Inspect skin for bruises, inflammation, edema, masses, tenderness, and sites of previous injections. • Use anatomic landmarks.	4. Injection site should be free of lesions. • Repeated daily injections should be rotated. • Avoids injury to underlying nerves, bone, or blood vessels.
5. Assist client into a comfortable position: • For vastus lateralis, lie flat or supine with knee slightly flexed. • For ventrogluteal, lie on side or back with knee and hip slightly flexed. •For dorsogluteal, lie prone with feet turned inward or on side with upper knee and hip flexed and placed in front of lower leg.	5.Relaxation minimizes discomfort. Distraction reduces anxiety.

• For deltoid, stand with arm relaxed at side or sit with lower arm relaxed on lap or lie flat with lower arm relaxed across abdomen. •Distract client by talking about an interesting subject.	
6.Use antiseptic swab to clean skin at site.	6. Circular motion and mechanical action of swab remove secretions containing microorganisms.
7.While holding swab between fingers of nondominant hand, pull cap from needle.	7.Swab remains accessible during procedure. Prevents contamination of needle
8. Administer injection: • Hold syringe between thumb and forefinger of dominant hand like a dart • Spread skin tightly, or pinch a generous section of tissue firmly for cachectic patients. • Inject needle quickly and firmly (like a dart) at a 90-degree angle. • Release the skin. • Grasp the lower end of the syringe with nondominant hand and position dominant hand to the end of the plunger. Do not move the syringe. • Pull back on the plunger to ascertain if needle is in a vein. If no blood appears, slowly inject the medication.	8. • Quick, smooth injection is easier with proper position of syringe. • Needle penetrates tight skin easier than loose skin. •Quick, firm injection minimizes discomfort. •Injection requires smooth manipulation of syringe parts. Movement of syringe may cause discomfort. • Aspiration of blood indicates intravenous placement of needle so procedure may have to be abandoned.

 Injection at the ventrogluteal site avoids major nerves and blood vessels.	
9. Quickly withdraw the needle while applying pressure with the antiseptic swab.	9. Supporting tissue around injection site minimizes discomfort.
10.Apply gentle pressure to site. Do not massage site. Apply bandage if needed. 	Massage damages underlying tissue.
11. Assist the client to a comfortable position.	11. Promotes comfort.
12. Discard the uncapped needle and syringe in a safe receptacle.	12. Decreases risk of needle stick.
13. Remove gloves and wash hands.	13. Reduces transmission of microorganisms.
14.Document the date, time, dose, route, site of injection, and signature or initials.	14. To obtain base line data

Procedure Checklist

Procedure	Mark	Trial 1	Trial 2	Trial 3	Comments
1. Wash hands and put on clean gloves.					
2. Prepare tray & take it to the patient's room					
3. Identify patient using two identifiers (e.g., name and birthday or name and account number) according to agency policy.					
4. Prepare the medication					
5. Draw the medicine					
6. Expel the air from the syringe					
7. Choose the site of injection (the site for intra-muscular)					
8. Using the iliac crest as the upper boundary divided the buttock into four. Clean the upper outer quadrant with alcohol swab:					
9. Stretch the skin and inject the medicine					
10. Draw back the piston (plunger) to check whether or not you are in the blood vessel (if blood returns, withdraw and get a new needle & re inject in a different spot)					
11. Push the drug slowly into the muscle					
12. When completed, withdraw the needle and massage the area with swab gently to and absorption.					
13. Place the patient comfortably					

14. Take care of the equipment you have used & return to their places 15. Chart the amount, time route and type of the medicine 16. Check the patient's reaction **Note:** 1. The needle for I.M. Injection should be long 2. Strict aseptic technique should be observed throughout the procedure. 3. Injection should not be given in areas such as inflamed, edematous, those containing moles and pus.					

4. Intravenous therapy

Administering Medications by Intravenous Bolus or Push Through an Intravenous Infusion

Definition: A medication can be administered as an IV bolus or push. This involves a single injection of a concentrated solution directly into an IV line. Drugs given by IV push are used for intermittent dosing or to treat emergencies. The drug is administered very slowly over at least 1 minute. This can be done manually or a syringe pump may be used.

Sites for IV injection

1. Dorsal Venous network
2. Dorsal metacarpal Veins
3. Cephalic Veins
4. Radial vein
5. Ulnar vein
6. Basilic vein
7. Median cubital vein
8. Greater saphenous vein

Purpose

- When the given drug is irritating to the body tissue if given through other routes.
- When quick action is desired.
- When it is particularly desirable to eliminate the variability of absorption.
- When blood drawing is needed (exsanguinations)

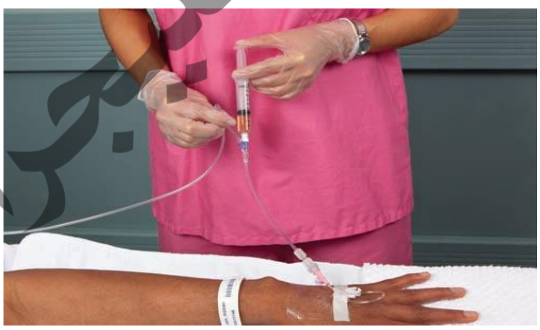
Equipment

- Antimicrobial swab
- Watch with second hand, or stopwatch
- Clean gloves
- Additional PPE, as indicated
- Prescribed medication
- Syringe with a needleless device or 23- to 25-gauge, 1-inch needle (follow facility policy)
- Syringe pump, if necessary
- Computer-generated Medication Administration Record (CMAR) or Medication Administration Record (MAR)

Procedure

Steps	Rational
1. Gather equipment. Check medication order against the original order in the medical record, according to facility policy. Check the patient's chart for allergies. Check the medication needs to be diluted before administration. Check the infusion rate	This comparison helps to identify errors that may have occurred when orders were transcribed.
2. Perform hand hygiene.	Hand hygiene prevents the spread of microorganisms.
3. Move the medication cart to the outside of the patient's room or prepare for administration in the medication area.	Organization facilitates error-free administration and saves time.

4. Prepare medication for one patient at a time.	This prevents errors in medication administration.
5. Recheck the label with the MAR before taking it to the patient.	This is a third check to ensure accuracy and to prevent errors
6. Transport medications and equipment to the patient's bedside carefully, and keep the medications in sight at all times.	Careful handling and close observation prevent accidental disarrangement of medications. Having equipment available saves time and facilitates performance of the task.
7. Ensure that the patient receives the medications at the correct time.	Check agency policy, which may allow for administration within a period of 30 minutes before or 30 minutes after designated time.
8. Perform hand hygiene and put on PPE, if indicated.	Hand hygiene and PPE prevent the spread of microorganisms. PPE is required based on transmission precautions
9. Identify the patient. Usually, the patient should be identified using two methods	Identifying the patient ensures the right patient receives the medications and helps prevent errors.
10. Close the door to the room or pull the bedside curtain.	This provides patient privacy.
11. Assess IV site for presence of inflammation or infiltration.	IV medication must be given directly into a vein for safe administration.

12. If IV infusion is being administered via an infusion pump, pause the pump.	Pausing prevents infusion of fluid during bolus administration and activation of pump occlusion alarms
13. Select injection port on tubing that is closest to venipuncture site. Clean port with antimicrobial swab.	Using port closest to needle insertion site minimizes dilution of medication. Cleaning deters entry of microorganisms when port is punctured.
14. Uncap syringe. Steady port with your nondominant hand while inserting syringe into center of port	This supports the injection port and lessens the risk for accidentally dislodging the IV or entering the port incorrectly.
15. Move your nondominant hand to the section of IV tubing just above the injection port. Fold the tubing between your fingers	This temporarily stops flow of gravity IV infusion and prevents medication from backing up tubing.
16. Pull back slightly on plunger just until blood appears in tubing	This ensures injection of medication into the bloodstream.
17. Inject the medication at the recommended rate 	This delivers correct amount of medication at proper interval according to manufacturer's directions.

18. Release the tubing. Remove the syringe. Do not recap the used needle, Discard the needle and syringe in the appropriate receptacle. Release the tubing and allow the IV fluid to flow.	Proper disposal of the needle prevents injury
19. Check IV fluid infusion rate. Restart infusion pump, if appropriate	Injection of bolus may alter rate of fluid infusion, if infusing by gravity.
20. Remove gloves and additional PPE, if used. Perform hand hygiene.	Removing PPE properly reduces the risk for infection transmission. Hand hygiene prevents the spread of microorganisms
21. Document the administration of the medication immediately after administration.	Timely documentation helps to ensure patient safety.
22. Evaluate the patient's response to medication within appropriate time frame.	The patient needs to be evaluated for therapeutic and adverse effects from the medication.

Procedure checklist

STEPS	Mark	Trial 1	Trial 2	Trial 3	comment
1.Gather equipment. Check medication order against the original order in the medical record, according to facility policy. Check the patient's chart for allergies. Check the medication needs to be diluted before administration. Check the infusion rate					
2.Perform hand hygiene.					
3.Move the medication cart to the outside of the patient's room or prepare for administration in the medication area.					
4.Prepare medication for one patient at a time.					
5.Recheck the label with the MAR before taking it to the patient.					
6.Transport medications and equipment to the patient's bedside.					
7.Ensure that the patient receives the medications at the correct time.					
8.Perform hand hygiene and put on PPE, if indicated.					
9.Identify the patient. using two methods					
10.Close the door to the room or pull the bedside curtain.					

11. Assess IV site for presence of inflammation or infiltration.					
12. If IV infusion is being administered via an infusion pump, pause the pump.					
13. Select injection port on tubing that is closest to venipuncture site. Clean port with antimicrobial swab.					
14. Uncap syringe. Steady port with your nondominant hand while inserting syringe into center of port					
15. Fold the tubing between your fingers					
16. Pull back slightly on plunger just until blood appears in tubing					
17. Inject the medication at the recommended rate					
18. Release the tubing. Remove the syringe. Do not recap the used needle, Discard the needle and syringe in the appropriate receptacle. Release the tubing and allow the IV fluid to flow.					
19. Check IV fluid infusion rate. Restart infusion pump, if appropriate					
20. Remove gloves and additional PPE, if used. Perform hand hygiene.					
21. Document the administration of the medication immediately after administration.					
22. Evaluate the patient’s response to medication within appropriate time frame.					

Collection of urine specimen

Objectives

Student will be able to

- Define collection of urine specimen
- identify nursing consideration for collection of urine specimen
- Determine equipment's needed during the collection of urine specimen
- Demonstrate the intervention for collection of urine specimen (midstream, catheter)
- Identify documentation for the collection of urine specimen

Collection of urine specimen (midstream) urine analysis and culture

Definition

Collecting a urine specimen for urinalysis and culture is an assessment measure to determine the characteristics of a patient's urine. A voided urine specimen for culture is collected midstream.

Nursing consideration:

1- Ask the patient about any medications that he or she is taking, because medications may affect the results of the test.
2- Assess for any signs and symptoms of a urinary tract infection, such as burning, pain (dysuria), or frequency.
3- Adequate amount of urine is obtained from the patient without contamination.

Equipment:

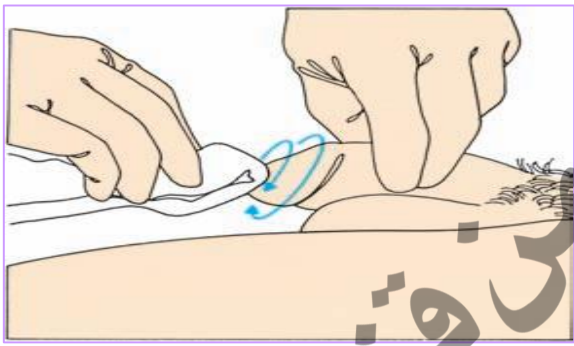
1. Moist cleansing towelettes or skin cleanser, water, and washcloth
2. Nonsterile gloves
3. Additional PPE, as Indicated

4. Sterile specimen container; urine collection tubes, based on facility policy
5. Biohazard bag
6. Appropriate label for specimen, based on facility policy and procedure

Procedure

Steps	Rational
1.Bring necessary equipment to the bedside stand or overbed table.	Bringing everything to the bedside conserves time and energy, Organization facilitates performance of tasks.
2.Perform hand hygiene and put-on PPE, if indicated.	Hand hygiene and PPE prevent the transmission of microorganisms. PPE is required based on transmission precautions.
3.Identify the patient. Explain the procedure to the patient. If the patient can perform the task without assistance after instruction, leave the container at bedside with instructions to call the nurse as soon as a specimen is produced.	Identifying the patient ensures the right patient receives the intervention and helps prevent errors. Explanation provides reassurance and promotes cooperation
4.Have the patient perform hand hygiene, if performing self-collection.	Hand hygiene prevents the transmission of microorganisms
5.Check the specimen label with the patient's identification bracelet. Label time specimen was collected route of collection.	Confirmation of patient identification information ensures the specimen is labeled correctly for the right patient.

6.Close curtains around bed and close the door to the room, if possible.	Closing the door or curtain provides for patient privacy.
7.Put on non-sterile gloves. Assist the patient to the bathroom, or onto the bedside bedpan. Instruct the patient not to defecate or discard toilet paper into the urine 	Gloves reduce the transmission of microorganisms. Stool and/or toilet paper may contaminate the specimen.
8.Instruct the female patient to separate the labia for cleaning of the area and during collection of urine. **<u>Female</u>** Opens the antiseptic towelettes provided in the prepackaged kit. If there is no kit, pours the antiseptic solution over the cotton balls. Spreads the labia and wipes down one side of the meatus from front to back using one pad and discards it. Then wipes the other side with a second pad. Wipes down the center over the urinary meatus with the third pad; then discards it. Maintains separation of the labia until the specimen is obtained.	Cleaning the perineal area or penis reduces the risk for contamination of the specimen. 

Male patients should use a towelette to clean the tip of the penis, wiping in a circular motion away from the urethra. Instruct the uncircumcised male patient to retract the foreskin before cleaning and during collection	
9.Have patient void a small amount of urine into the toilet, bedpan, or commode. The patient should then stop urinating briefly, then void into collection container. Collect specimen (10 to 20 mL is sufficient), and then finish voiding. Do not touch the inside of the container or the lid.	Collecting a midstream specimen ensures that fresh urine is analyzed. Some urine may have collected in the urethra from the last void. By voiding a little before collecting the specimen, the specimen will contain only fresh urine.
10. Carefully replaces the sterile container lid, touching only the outside of the cap and container.	Placing the lid on the container helps to keep the specimen clean and prevents spills
11. Assist the patient from the bathroom, or off the bedpan. Provide perineal care, if necessary.	Perineal care promotes patient comfort and hygiene.
12. Remove gloves and perform hand hygiene.	reduces the risk for infection transmission and contamination of other items.
13. Place label on the container per facility policy. Place container in plastic, sealable biohazard bag.	Proper labeling ensures accurate reporting of results. Packaging the specimen in a biohazard bag prevents the person transporting the container from coming in contact with urine.

	
14. Remove other PPE, if used. Perform hand hygiene.	reduces the risk for infection transmission and contamination of other items.
15. Transport the specimen to the laboratory as soon as possible. If unable to take the specimen to the laboratory immediately, refrigerate it.	If not refrigerated immediately, urine may act as a culture medium, allowing bacteria to multiply and skewing the results of testing.

Procedure Check List

STEPS	Mark	Trial 1	Trial 2	Trial 3	comment
1. Bring necessary equipment to the bedside stand or overbed table.					
2. Perform hand hygiene and put on PPE, if indicated					
3. Identify the patient. Explain the procedure to the patient. If the patient can perform the task without assistance after instruction, leave the container at bedside with instructions to call the nurse as soon as a specimen is produced					
4. Have the patient perform hand hygiene, if performing self-collection.					
5. Check the specimen label with the patient's identification bracelet. Label time specimen was collected route of collection.					
6. Close curtains around bed and close the door to the room, if possible.					
7. Put on unsterile gloves. Assist the patient to the bathroom, or onto the bedside commode or bedpan. Instruct the patient not to defecate or discard toilet paper into the urine.					
8. Instruct the female patient to separate the labia for cleaning of the area and during collection of					

urine. Male patients should use a towelette to clean the tip of the penis, wiping in a circular motion away from the urethra.					
9. Have patient void a small amount of urine into the toilet, bedpan, or commode. The patient should then stop urinating briefly, then void into collection container. Collect specimen (10 to 20 mL is sufficient), and then finish voiding. Do not touch the inside of the container or the lid.					
10. Carefully replaces the sterile container lid, touching only the outside of the cap and container.					
11. Assist the patient from the bathroom, or off the bedpan. Provide perineal care, if necessary.					
12. Remove gloves and perform hand hygiene.					
13. Place label on the container per facility policy. Place container in plastic, sealable biohazard bag.					
14. Remove other PPE, if used. Perform hand hygiene					
15. Transport the specimen to the laboratory as soon as possible. If unable to take the specimen to the laboratory immediately, refrigerate it.					

Collection of urine specimen (indwelling urinary catheter) urine analysis &culture

Indwelling catheter drainage tubes have special sampling ports in the tubing for removal of urine for testing. Most sampling ports are needleless systems. However, some ports require the use of a needle to access the sampling port.

Nursing consideration

1- Do not open the drainage system to obtain urine specimens to avoid contamination of the system and bladder infection. Never take urine specimens from the catheter drainage bag because the urine is not fresh and bacteria may be present on the bag
2- Assess the characteristics of the urine draining from the catheter.
3- Inspect the catheter tubing to identify the type of sampling port present
4- Slowly inject urine into the specimen container. Take care to avoid touching the syringe tip to any surface. Do not touch the edge or inside of the collection container.

Equipment

1. 10-mL sterile syringe
2. Blunt cannula or 18-gauge needle, if needed, based on specific catheter in use
3. Alcohol or other disinfectant wipes
4. Nonsterile gloves
5. Additional PPE, as indicated
6. Sterile specimen container; urine collection tubes, based on facility policy
7. Biohazard bag
8. Appropriate label for specimen, based on facility policy and procedure.

Procedure

Steps	Rational
1. Bring necessary equipment to the bedside stand or overbed table.	Bringing everything to the bedside conserves time and energy, Organization facilitates performance of tasks.
2. Perform hand hygiene and put-on PPE, if indicated.	Hand hygiene and PPE prevent the transmission of microorganisms. PPE is required based on transmission precautions.
3. Identify the patient. Explain the procedure to the patient.	Identifying the patient ensures the right patient receives the intervention and helps prevent errors. Explanation provides reassurance and promotes cooperation
4. Check the specimen label with the patient's identification bracelet. Label time specimen was collected route of collection.	Confirmation of patient identification information ensures the specimen is labeled correctly for the right patient.
5. Close curtains around bed and close the door to the room, if possible	Closing the door or curtain provides for patient privacy.
6. Put on unsterile gloves.	Gloves reduce the transmission of microorganisms
7. Clamp the catheter drainage tubing or bend it back on itself distal to the port. If an insufficient amount of urine is present in the tubing, Allow the tubing to remain	Clamping the tubing ensures the collection of an adequate amount of fresh urine. Clamping for an extended period of time leads to overdistention

clamped up to 30 minutes, to collect a sufficient amount of urine, unless contraindicated. 	of the bladder. Clamping may be contraindicated based on the patient's condition.
8. Cleanse aspiration port with alcohol wipe and allow port to air dry.	Cleaning with alcohol deters entry of microorganisms when the needle punctures the port.
9. Attach the syringe to the needleless port or insert the needle into the port. Slowly aspirate enough urine for specimen (usually 10 mL is Adequate) 	Using a blunt-tipped needle prevents a needlestick. Collecting urine from the port ensures that the specimen will contain fresh urine. Unclamping catheter drainage tubing prevents overdistention and injury to the patient's bladder
10. If a needle or blunt-tipped cannula was used on the syringe, remove from the syringe before emptying the urine from the syringe into the specimen cup. Place the needle into sharps collection container. Slowly inject	Forcing urine through the needle breaks up cells and impedes accurate results of microscopic urinalysis. Safe disposal of sharps prevents accidental injury. If the urine is injected quickly

urine into specimen container. Replace lid on container. Dispose of syringe appropriately.	into the container, it may splash out of the container or into the nurse’s eyes
11. Remove gloves and perform hand hygiene	Removing gloves properly reduces the risk for infection transmission and contamination of other items. Hand hygiene reduces the transmission of microorganisms.
12. Place label on the container per facility policy. Place container in plastic sealable biohazard bag. 	Proper labeling ensures accurate reporting of results. Packaging the specimen in a biohazard bag prevents the person transporting the container from coming in contact with the specimen.
13. Remove other PPE, if used. Perform hand hygiene.	Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene reduces the transmission of microorganisms.
14. Transport the specimen to the laboratory as soon as possible. If unable to take the specimen to laboratory immediately, refrigerate it	If not refrigerated immediately, urine may act as a culture medium, allowing bacteria to multiply and skewing the results of testing.

Procedure Check List

Steps	Mark	Trial 1	Trial 2	Trial 3	comment
1. Bring necessary equipment to the bedside stand or overbed table.					
2. Perform hand hygiene and put on PPE, if indicated					
3. Identify the patient. Explain the procedure to the patient.					
4. Check the specimen label with the patient's identification bracelet. Label time specimen was collected route of collection.					
5. Close curtains around bed and close the door to the room, if possible					
6. Put on unsterile gloves.					
7. Clamp the catheter drainage tubing to sufficient amount of urine or remain clamped up to 30 minutes.					
8. Cleanse aspiration port with alcohol wipe and allow port to air dry.					
9. Attach the syringe to the needleless port or insert the needle into the port. Slowly aspirate enough urine for specimen (usually 10 mL is Adequate)					
10. If a needle was used on the syringe, remove from the syringe before emptying the urine from					

the syringe into the specimen cup. Place the needle into a sharps collection container.					
11. Place the needle into sharps collection container. Slowly inject urine into specimen container. Replace lid on container. Dispose of syringe appropriately.					
12. Remove gloves and perform hand hygiene.					
13. Place label on the container per facility policy. Place container in plastic, sealable biohazard bag.					
14. Remove other PPE, if used. Perform hand hygiene					
15. Transport the specimen to the laboratory as soon as possible. If unable to take the specimen to the laboratory immediately, refrigerate it.					

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